STITCH: ROBUST APPEARANCE-INVARIANT VISUAL PLACE RECOGNITION FOR ANCHORING SOCIAL AR CONTENT IN THE REAL WORLD

A DISSERTATION

by

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November, 2026

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Candidate's Declaration

I, **PRONOB BHATTACHARJEE**, present the dissertation entitled "**Stitch: Robust Appearance-Invariant Visual Place Recognition for Anchoring Social AR Content in the Real World**" in partial fulfilment of the requirements for the award of the degree of **MASTER OF TECHNOLOGY** in Artificial Intelligence, submitted to the Mehta Family School of Data Science and Artificial Intelligence, Indian Institute of Technology Roorkee. The report documents the implementation, executed experiments and attributed prior research for the Stitch project, prepared under the guidance of Dr. R. Balasubramanian.

All experimental results are linked to the accompanying commands, datasets and software artifacts. Published methods, public datasets and automated assistance are acknowledged in the report. This declaration is prepared for review and signature by the candidate.

Date: 6 November 2026 (PRONOB BHATTACHARJEE)
Place: Roorkee

CERTIFICATION

The dissertation submitted by PRONOB BHATTACHARJEE (Enrollment No. 25565017) is presented for review under the guidance of Dr. R. Balasubramanian. Final certification and acceptance are to be completed by the supervisor and the institution after review.

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Signature: _____

Dedicated to

my family, teachers

and mentors

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The report and presentation adopt the institutional artwork and layout references supplied by the author. The experiments, scripts and provenance records accompany the dissertation so that its claims can be reviewed and reproduced. Automated implementation and writing assistance formed part of artifact preparation; the author is responsible for reviewing the final submission.

I am sincerely grateful to my supervisor, Dr. R. Balasubramanian, for the guidance, encouragement and feedback provided throughout this internship. I also thank my family and friends for their constant support.

Abstract

Stitch proposes a social augmented-reality interface in which digital posts are associated with physical places and recovered when a later visitor observes the same scene. Its original implementation matches SIFT features to one reference image and projects content using a planar homography. This thesis investigates whether pretrained representations improve recognition and anchoring under appearance change while retaining a reproducible path toward mobile deployment.

A modular implementation compares SIFT with FLANN, SuperPoint with LightGlue, CosPlace global retrieval, and a hierarchical combination of retrieval, local matching, geometric verification, and rendering. Experiments use a documented HPatches subset, a custom cross-time subset of real Tokyo 24/7 query photographs, and seeded photometric transformations of repository and benchmark images. Original scripts are executed through a bounded file-input adapter. All reported measurements, plots and tables are derived from preserved run outputs.

The evaluation distinguishes retrieval, geometric acceptance and ground-truth homography accuracy. Tokyo metadata supports place labels but does not establish metric camera pose or planar anchoring error. Short contrastive adaptation trains only a projection head on disjoint HPatches places. Model export and integer quantisation are tested using actual ONNX inference. The report documents accuracy losses, rejected anchors, and the limits of narrow galleries and synthetic illumination changes.

The resulting artifact demonstrates a reproducible planar-AR research pipeline. It does not establish a city-scale AR cloud, a six-degree-of-freedom localisation benchmark result, or phone-device performance. The deployment analysis specifies the mapping, calibration, tracking, privacy and evaluation work needed before persistent public AR content can be supported responsibly.

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Nomenclature

$H_{r \to q}$	Homography mapping reference pixels to query pixels
AR	Augmented reality
CI	Confidence interval
CNN	Convolutional neural network
FLANN	Fast library for approximate nearest neighbours
GeM	Generalised mean pooling
GPS	Global positioning system
ONNX	Open Neural Network Exchange
PnP	Perspective-n-point
QDQ	Quantise–dequantise graph representation
RANSAC	Random sample consensus
RSS	Resident set size
SfM	Structure from motion
SIFT	Scale-invariant feature transform
SLAM	Simultaneous localisation and mapping
VPR	Visual place recognition

Chapter 1

Introduction

1.1 Motivation and application context

Stitch begins with an application idea: a social feed whose organising surface is the physical world. A visitor could leave a photograph on a building facade, attach a note to a meeting point, or place a virtual object beside a familiar landmark. A subsequent visitor would recover that content by pointing a camera at the same scene. The repository describes this ambition through the metaphor of a social wall distributed across real surroundings [1]. This thesis uses "Facebook in the real world" as a motivating description, rather than as a statement of affiliation or of implemented social-network functionality.

The idea creates a correspondence problem before it creates a graphics problem. The system must decide which stored place, if any, is visible. It must then determine how content associated with that place should project into the current camera image. A visually similar building is insufficient evidence for rendering a post on a particular window. Likewise, retrieving the correct street does not identify a stable coordinate frame for a three-dimensional object. Recognition and geometric alignment therefore require separate outputs and separate evaluation criteria.

The original software provides a compact demonstration of this distinction. It detects SIFT features in a reference image and a camera frame, matches their descriptors, estimates a homography, and warps an overlay into the frame. The latest script adds optical-flow tracking and smoothing. These operations are useful for a locally planar target already represented by a reference photograph. They do not by themselves provide a place database, a city-scale search index, or a persistent world coordinate system. The engineering task is to extend that working demonstration while retaining its inspectable geometric basis.

An appearance change challenges the first correspondence step. Daylight produces different shadows, contrast and visible detail from artificial illumination at night. Seasonal vegetation may remove or reveal structures. Weather can introduce reflections or reduce contrast. These changes interact with viewpoint: a correctly recognised scene may have little shared visible surface with the reference. Long-term visual-localisation benchmarks explicitly examine these difficulties [2]. A method suitable for Stitch must therefore be assessed on more than an unchanged photograph or a visually convincing overlay video.

1.2 Problem statement

Let a place database contain reference images $\mathcal{D} = \{I_i\}$, place identifiers p_i, and content records $\mathcal{C}_{p_i}$. Given a query image I_q, the system should return a place hypothesis, evidence supporting that hypothesis, a geometric transformation where identifiable, and the associated content. A rejection is a valid result when the evidence is inadequate. The planar research prototype studied here returns

$$\mathcal{F}(I_q, \mathcal{D}) = (\widehat{p}, \widehat{H}_{r \to q}, \mathcal{C}_{\widehat{p}}, s), \tag{1.1}$$

where s contains diagnostic matching and verification information. When no candidate passes verification, $\widehat{p}$ and $\widehat{H}$ are absent and no anchored content is rendered.

The desired invariance is conditional. A representation should ignore exposure and other nuisance changes while retaining evidence that distinguishes places. It cannot discard all local structure and still determine the intended surface. Nor can it restore image information destroyed by underexposure. "Appearance-invariant" in the project title therefore identifies the research objective; it does not claim mathematical invariance or universal robustness. The experiments measure how particular pretrained representations behave under documented changes.

The resource constraint is also explicit. The available host is a laptop with an Apple M2 processor and a limited-memory configuration, recorded in the environment artifact. The implementation uses CPU inference because the recorded runtime did not expose an available accelerator. A mobile-class design is investigated through compact descriptors, bounded feature counts, model export and quantisation. A laptop CPU timing cannot establish performance on a phone, and the thesis does not make that substitution.

1.3 Objectives

The first objective is a reproducible classical reference. The original scripts must be inspected and executed on their bundled video without requiring an unavailable private camera stream. A refactored implementation should retain the SIFT–FLANN–RANSAC structure while making image coordinates, failures and rendering masks explicit. This baseline anchors the discussion in measured behaviour rather than an assumed weakness of handcrafted features.

The second objective is a controlled comparison of learned components. SuperPoint with LightGlue supplies local correspondences; CosPlace supplies an image-level descriptor. Each has a different role. Their combination is evaluated alongside exhaustive local matching and global-only retrieval. Identical query lists and geometry settings are used within each protocol so that a benefit is not explained by an easier test subset or a more permissive RANSAC threshold.

The third objective is to connect recognition to content. A place record must support reference photographs and posted assets, and a query must retrieve the content associated with its verified place. The implemented output is an image overlay attached to a reference-plane polygon. Video capture and rendering are included in the demonstration. The more ambitious multi-user service and three-dimensional anchor lifecycle are specified as architectural extensions,

with implementation boundaries stated explicitly.

The fourth objective is a thesis whose evidence can be audited. Raw predictions and pair statistics are retained. Plotting and table generation consume those files, and prose-level headline values use generated LaTeX macros. Dataset and checkpoint hashes identify the inputs. Commands and exit codes identify the actions that produced results, including failed download and environment checks. The report should make an unmeasured quantity visibly unavailable rather than filling a requested table with an invented estimate.

1.4 Research questions

1. Does pretrained local matching improve correspondence and accepted homography accuracy under the measured illumination and viewpoint changes?
2. How does global retrieval affect the candidate search needed for reliable place recognition, and when does its shortlist discard a geometrically useful reference?
3. How often does a plausible retrieval result lead to an accepted and correct place, and how does that differ from ground-truth planar alignment accuracy?
4. Does short contrastive adaptation of a compact projection head improve retrieval on held-out places and real cross-time images?
5. What accuracy, size and latency trade-offs follow from an executed descriptor export and INT8 quantisation attempt?

Each question maps to an observable artifact. Pair statistics address the first; ranked predictions and candidate-count ablations address the second and third; the seeded training records address the fourth; native-versus-ONNX descriptors and timed inference address the fifth. Metric six-degree-of-freedom localisation is excluded from these answers because the downloaded evaluation subsets do not supply the required calibrated poses and three-dimensional map.

1.5 Scope of the empirical contribution

The public-data contribution comprises a deterministic HPatches subset and a custom Tokyo cross-time protocol. HPatches contributes real illumination and viewpoint changes with image-to-image homographies [3]. Tokyo contributes photographs captured at different times with place and orientation labels [4]. The latter protocol uses a small reference gallery drawn from the query collection; it is deliberately named a cross-time subset rather than the official Tokyo street-view retrieval benchmark. Its results support a narrow comparison under matched labels, not a claim of benchmark-leading performance.

Synthetic queries serve a different purpose. They provide known geometric maps while varying brightness, gamma, colour balance, noise and contrast. Since all transformations are seeded, the same source identity can be followed through a controlled perturbation. However, altered

pixels retain the original scene, visibility and much of its texture. Their performance is therefore interpreted as a stress test of an implementation, not as a substitute for real night imagery or evidence about winter conditions.

The methodological contribution is the integration and evaluation of available pretrained components in the Stitch setting. No new foundation model, feature detector, or neural matcher is claimed. The short training experiment changes only a projection head on frozen global descriptors. Its purpose is to expose the practical consequences of limited adaptation data and descriptor compression. Treating a neutral or negative outcome as a valid answer is necessary to avoid selecting a favourable training seed after observing the test set.

The software contribution includes modular extraction, matching, geometry, rendering, database and I/O modules; a command-line interface; configuration and tests; data/model acquisition scripts; evaluation and demonstration code; and a generated-report workflow. The reference database stores content alongside place identifiers so that the experiment exercises retrieval of an actual posted asset rather than merely drawing an arbitrary polygon after matching. Persisting that record on disk establishes local repeatability, while cross-device consistency remains future work.

1.6 Assumptions and boundaries

The reference photograph is assumed to represent a surface on which a planar overlay is meaningful. Within that model, an estimated homography maps reference pixels to query pixels. A street scene is generally not planar, so an accepted Tokyo homography is evidence of a consistent image region, not proof that every part of the street obeys the same map. This distinction motivates the separate reporting of geometric acceptance and true homography success.

The gallery is treated as known and locally available during the experiment. Reference descriptors and local features may be cached before a query arrives. Timings state whether they include this setup, image decoding, rendering and content-file access. The production design would need to constrain cache size and fetch latency. Those costs are analysed symbolically and through measured component sizes; they are not represented as measured cloud-service costs.

The work assumes that dataset access does not imply unrestricted redistribution or commercial permission. Software and imagery have separate provenance. Official documentation is retained where available, and manual-fetch instructions identify larger benchmarks whose image and mapping archives were not downloaded. A future product must resolve additional content rights, user consent and location privacy before reusing a research pipeline in public spaces.

1.7 Organisation of the thesis

Chapter 2 reviews the representations, geometric models, datasets and anchor services relevant to the project. Chapter 3 analyses the original scripts and their recorded execution. Chapter 4 describes the implemented modules and the proposed multi-user service boundary. Chapter 5

defines the downloaded subsets and all preparation rules. Chapter 6 develops the mathematical formulation and algorithms. Chapter 7 specifies hardware, metrics and experimental controls. Chapter 8 reports the measured comparisons and validity threats. Chapter 9 examines export, quantisation and scaling. Chapter 10 treats privacy and abuse as system requirements. Chapter 11 answers the research questions and defines a staged roadmap. The appendix provides reproduction commands.

Chapter 2

Background and Literature Review

2.1 Recognition, localisation and anchoring

Visual place recognition asks which previously observed place is depicted in a query. Image retrieval implements this as ranking a gallery using visual descriptors. A retrieved image can supply an approximate location or a candidate region, but its similarity score does not determine the query camera's geometric pose. Visual localisation instead estimates the camera transformation relative to a map. In a calibrated three-dimensional setting that transformation has rotation and translation components. The established long-term benchmark distinguishes retrieval-like recognition from six-degree-of-freedom localisation under changing conditions [2].

Anchoring adds an application coordinate convention. A post needs a position or surface in the map, and the current device needs a transformation from that map to its view. A planar reference image can provide such a convention for image overlays: content coordinates map to reference-image pixels and then to query-image pixels. General three-dimensional objects require a world or surface coordinate frame whose scale and orientation are known. These outputs impose different observability requirements and should not share a single undifferentiated "localisation success" score.

This separation guides the experimental design. Global-only retrieval is evaluated as retrieval. A local matcher is evaluated on image pairs whose relationship is known independently. A complete candidate-search pipeline is evaluated both for its ranked place result and for whether its selected candidate passes verification. On HPatches, an additional ground-truth comparison checks the estimated mapping. This arrangement permits a matcher to be good even when retrieval misses its correct candidate, and permits a retrieval method to be good even when a query lacks enough shared surface for anchoring.

2.2 Handcrafted local features

2.2.1 SIFT

SIFT constructs local descriptors around characteristic image structures selected across scale. Orientation assignment and gradient-distribution normalisation provide robustness to several nuisance transformations. Lowe's account also makes clear that robustness is bounded: perspective distortion, severe appearance change and absent detail remain relevant [5]. Stitch uses the OpenCV implementation rather than reimplementing the detector.

For this application, a local descriptor provides more than an image-level match. Each descriptor remains associated with a pixel location, so accepted correspondences constrain a geometric map. This is the central reason a global embedding cannot directly replace the original pipeline. The comparison tests whether learned keypoints and descriptors provide better correspondences while preserving that association.

2.2.2 Approximate matching and the ratio test

FLANN selects approximate nearest neighbours using indexing strategies suited to descriptor search [6]. In the upstream scripts, a randomised tree index and a bounded search effort retrieve the first two neighbours for each reference descriptor. A ratio criterion rejects a match when its best neighbour is insufficiently separated from the second-best candidate. This ambiguity test is useful for repeated textures, although it cannot establish physical scene identity by itself.

The implementation must also handle cases omitted by compact demonstrations. A query with no descriptors produces no candidate correspondences. A reference descriptor may receive fewer than two neighbours if the target set is too small. Neither condition should raise an unhandled tuple-unpacking error or lead to a reused anchor being interpreted as a new recognition. The refactor makes empty sets and rejection first-class outputs.

2.3 Learned local representations and matching

2.3.1 SuperPoint

SuperPoint jointly predicts interest points and descriptors from an image. Its training framework uses synthetic supervision and homographic adaptation to improve repeatability without requiring manually annotated natural-image keypoints [7]. The present work uses publicly released pretrained weights. Neither the detector nor its descriptor backbone is trained on the evaluation subset.

The practical hypothesis is that joint learning may preserve useful correspondence evidence when raw contrast or local gradient distributions change. That hypothesis remains empirical. The learned extractor can still miss features in severe low light, repeated patterns or regions with little texture. A bounded keypoint count is used for both local methods, so the learned method is not granted an unlimited correspondence budget.

2.3.2 LightGlue

LightGlue revisits sparse learned matching with adaptive computation. Its official implementation supports SuperPoint and several other feature families and exposes mechanisms that adjust work to pair difficulty [8, 9]. The experiments use SuperPoint-compatible pretrained weights, CPU inference and the recorded confidence settings. Flash-attention acceleration is disabled so the tested execution path is explicit.

Adaptivity matters for Stitch because a difficult night query may require more work than an easy repeated view. A single average latency can conceal that variation. The artifact consequently records per-pair times and separately profiles complete queries. However, a laptop inference result does not demonstrate the same adaptive performance in a mobile runtime: export operators, memory allocation and backend support must also be verified.

2.4 Global descriptors for place retrieval

2.4.1 NetVLAD

NetVLAD makes residual aggregation trainable and uses weak place supervision to learn image embeddings for retrieval [10]. Its broader lesson is that visual similarity can be learned from place relationships rather than from generic object categories alone. In Stitch, this role is coarse candidate selection; it reduces the number of local reference-query comparisons before geometric verification.

2.4.2 CosPlace

CosPlace addresses large-scale visual geo-localisation through a classification-based training formulation and releases compact pretrained retrieval models [11, 12]. The artifact selects a ResNet18 model with a 512-dimensional output. The choice provides a smaller encoder than the larger released backbones and a straightforward convolutional export path. It is an engineering selection for the available host, rather than a demonstrated optimum among all modern retrieval systems.

2.4.3 Candidate retrieval as a bottleneck

If a correct reference is absent from the selected shortlist, a downstream matcher cannot recover it without expanding the search. A small shortlist can therefore improve latency while reducing the ceiling on successful anchoring. Increasing its size may recover relevant references but also adds expensive nonmatching pairs and more opportunities for a false geometric acceptance. The stage ablation varies shortlist size on exactly the same queries to expose this trade-off.

A retrieval score also has no universal probability interpretation. Cosine similarity measures proximity in the selected descriptor space. Its distribution changes with gallery composition, camera conditions and training domain. Treating a high similarity as permission to display content would bypass the local evidence needed to identify an anchoring surface. The no-geometry ablation is retained as a diagnostic, not as the recommended rendering policy.

2.5 Projective geometry and robust estimation

Image correspondences constrain different geometric models depending on scene structure and calibration. Homographies describe plane-induced image mappings and pure-rotation cases. Essential matrices describe calibrated two-view epipolar relationships. PnP estimates camera pose from known three-dimensional points and their image projections [13, 14]. Stitch's available assets and HPatches supply a direct planar formulation; the downloaded Tokyo metadata does not supply the calibrated 2D–3D relationships needed for PnP.

RANSAC estimates a model from subsets of tentative correspondences while rejecting gross outliers [15]. It does not transform an inappropriate model into a physically correct one. A street scene may admit many correspondences on one facade while foreground objects remain inconsistent with the same homography. Similarly, repeated windows may support a projective map to the wrong location. Inlier support is therefore necessary evidence for this prototype, but it is not complete evidence of place identity or three-dimensional consistency.

2.6 Hierarchical visual localisation

Coarse-to-fine localisation combines global hypotheses with local geometric evidence. HF-Net demonstrates this principle using jointly predicted local and global representations [16]. The present system adopts the hierarchy but uses separate pretrained networks and a planar verifier. Its proposed combination is an integration choice, not a claim to originate hierarchical localisation.

Separating the stages remains valuable for a small research artifact. Each component can be replaced and each failure can be attributed to a recorded intermediate output. A wrong shortlist, an inadequate correspondence set and an inaccurate accepted homography become different diagnoses. The resulting modularity also exposes costs that a monolithic end-to-end timing would hide, such as reference extraction and content asset loading.

2.7 Benchmarks and domain coverage

HPatches offers image sequences with illumination or viewpoint variation and known planar relationships [3, 17]. It is suited to evaluating image alignment, but it does not establish geographic recognition over a city or metric pose accuracy. A small distractor gallery assembled from its sequence references is an explicit custom retrieval task.

Tokyo 24/7 was designed around changes in time of day and viewpoint, with camera-phone queries and a street-view database [4]. Using its accessible query photographs as both cross-time references and queries creates a useful paired recognition experiment but changes the task. The benchmark's original database scale and ground-truth conventions must not be carried over to this subset's results.

Aachen Day-Night, RobotCar Seasons and Extended CMU Seasons extend evaluation toward long-term metric localisation [2, 18]. MSLS provides broader geographic, seasonal and sequence variation [19]. Those capabilities motivate future validation, while their absence from the exe-

cuted image set limits what this thesis can conclude about seasonal robustness and map-level localisation.

2.8 Persistent AR services and lifecycle

ARCore Cloud Anchors uses hosted scene information and an anchor identifier to support later resolution by devices observing the environment [20]. ARKit world maps support saving spatial state and attempting relocalisation in a later session [21]. These systems demonstrate that persistence involves mapping and coordinate recovery, not only content storage. Their documentation informs the proposed integration boundary; neither service is called by the experiments.

Azure Spatial Anchors is relevant historically, but Microsoft states that the service was retired in November 2024 [22]. It is therefore discussed as background rather than recommended as a new dependency. This distinction illustrates why service status must be checked independently of the publication date of an AR architecture proposal.

2.9 Research gap addressed by the artifact

The literature already provides strong representations, matchers and hierarchical localisation designs. The immediate gap in Stitch is the absence of an evaluated connection between those components and its reference-plane rendering demonstration. The artifact addresses that gap by preserving the classical behaviour, introducing explicit place records, executing controlled comparisons, and making rejection and uncertainty visible. The remaining gap is larger: metric map construction, cross-device anchor management and realistic phone validation are still required for the original social-AR ambition.

Chapter 3

Analysis of the Existing System

3.1 Inspected source snapshot

The upstream repository was cloned and inspected at the revision recorded in the bibliography and model/data provenance. Its executable Python content comprises `tep.py`, `tep2.py` and `tep3.py`. Bundled media include a reference image, a test image, an overlay, a short video and an animated walkthrough. No labelled place database, learning code or downloadable benchmark is included [1]. The original scripts remain present and unmodified in the working artifact.

The inspection distinguishes what is implied by comments from what the code actually does. In particular, the latest script is not simply a frame-by-frame SIFT matcher: it includes optical flow, adaptive detection intervals, forward–backward checking and homography smoothing. Consequently, comparing its tracking speed directly with a learned re-recognition pass would confound different workloads. Both workloads are useful, but they answer different questions.

3.2 The file-processing script

3.2.1 Initialisation

`tep.py` reads `reference.jpg` in grayscale and `overlay.png` in colour. It constructs a SIFT extractor with the upstream feature budget, extracts reference descriptors once, and resizes the overlay to the reference dimensions. A FLANN matcher uses the recorded tree and search settings. The input is `video.mp4`; the output writer uses a reduced playback frame rate. That output frame rate is a presentation choice and must not be interpreted as measured inference throughput.

Reference caching is an important strength. It avoids repeating work that is unchanged across frames. The same principle is preserved in the modular system, where local reference features and global gallery descriptors are prepared before an online query. A fair runtime report must therefore state whether this preparation is included. The upstream bounded execution includes initialisation, whereas the later warm profiles isolate query-time processing.

3.2.2 Frame processing and verification

Each frame is converted to grayscale, blurred and enhanced with CLAHE. SIFT descriptors are extracted from the enhanced image. The first two descriptor neighbours are compared with the script's ratio threshold. A tentative homography is estimated only after enough matches survive; it is used only after an additional inlier-count test. These operations are conceptually sensible but encode thresholds and preprocessing directly in a monolithic script.

The reference and query are not processed identically. The reference is loaded directly as grayscale, whereas the query receives blur and local contrast enhancement. This asymmetry is part of the preserved upstream behaviour. The controlled A–D experiment instead uses a common image-resizing convention and shared geometry thresholds, with the difference documented. The latter is a normalised comparison of methods, not a claim of byte-for-byte reproduction of every upstream preprocessing decision.

3.2.3 Persistence and compositing

When a new homography is unavailable, `tep.py` can reuse the previous one for a bounded number of frames. This can maintain visual continuity during a short detection failure. It can also display an old transformation after camera motion, because the reused map is not updated by a geometric motion model. A rendered frame is therefore not equivalent to a successful fresh recognition.

The overlay mask is inferred from warped image intensity. Pixels that are sufficiently dark are treated as transparent. This is simple for demonstration assets, but it makes legitimate black content disappear and does not represent semi-transparency. The refactored renderer uses an explicit alpha plane and premultiplied colour. A unit test verifies that an opaque black overlay remains opaque and that a fully transparent one leaves the frame intact.

3.3 The first live-stream variant

`tep2.py` follows the same broad matching and rendering sequence but reads a stream and displays a window. In the inspected snapshot, the example URL definition is commented out while the variable is subsequently used. Native execution therefore lacks a required input value. The script also continues when a frame read fails, without a finite retry policy or an EOF distinction.

These issues are I/O failures rather than evidence about the SIFT algorithm. The execution adapter supplies a file source and suppresses display calls so that the vision code can be exercised in this environment. It does not invent a live phone experiment. The code path is labelled "bounded file-input adapter" in the result record, and the original stream requirement is retained in the source analysis.

3.4 The latest detection-and-tracking variant

3.4.1 Detection and scheduling

`tep3.py` reads a hard-coded private DroidCam URL and resizes frames to a fixed working shape. During detection it downsamples the grayscale query, extracts SIFT descriptors, matches them to the full reference descriptors, and rescales matched query points back to working-frame coordinates. This coordinate correction is necessary: fitting the homography in a mixture of downsampled and full-frame pixels would corrupt the overlay location.

Detection is attempted at an interval rather than on every frame. Failed detections increase that interval up to a cap. A successful detection resets the schedule and initialises tracking. This reduces repeated expensive searches when a target is absent, but it also introduces recognition delay. A frame-count average can look favourable partly because some frames perform little or no recognition work. The bounded execution therefore reports decoded frames, homography calls and warp calls separately.

3.4.2 Optical flow and consistency checks

Once points are available, pyramidal Lucas–Kanade optical flow propagates their positions between frames. A periodic backward pass checks whether tracking approximately returns to the previous point positions. A geometric filter further rejects motions that differ substantially from median motion. These checks target local tracking errors; they do not compare a newly visible place with a database.

The source contains implementation limitations worth preserving in the audit. It does not fully validate all optional optical-flow outputs before indexing them, and the backward-status array is not incorporated into every acceptance decision. A feature set may therefore be lost abruptly rather than producing a clean diagnostic rejection. The modular recognition path handles empty feature sets and model-estimation failures explicitly. It does not claim that every tracking behaviour has been ported or improved.

3.4.3 Homography smoothing

The tracking correspondences estimate a fresh homography. A determinant-based validity check rejects nonfinite or implausibly scaled matrices. The script may lock the prior transformation when average point motion is small, or blend matrix entries with the new estimate when motion is larger. This can reduce visible jitter in the demonstration.

However, a homography is defined only up to scale. Entrywise averaging is not generally invariant to that representation, and a smooth matrix sequence need not correspond to the intended surface motion. The determinant is also affected by multiplying the whole matrix by a scalar. The new verifier normalises the matrix representation, checks rank and numerical conditioning, and examines the projected reference boundary. More principled temporal filtering in a calibrated pose or surface-coordinate space remains a deployment extension.

3.5 Execution protocol and measured baseline

The three scripts were executed through `scripts/run_originals.py`. The adapter redirects video capture to the bundled file, supplies the missing URL variable where needed, suppresses GUI display, and terminates the run after the same decoded-frame budget. It wraps homography and warping calls to count them without changing their arguments or results. The exact command, elapsed time, exit state and raw output are preserved in the command journal.

Table 3.1: Measured upstream-script execution on the same bounded file input. Timing includes initialisation and file I/O.

Script	Frames	H estimates	Warps	ms/frame
tep.py	90	90	90	55.70
tep2.py	90	90	90	56.45
tep3.py	90	87	87	11.31

Table 3.1 reports the recorded measurements. The latest script's lower average time is consistent with its design: many frames track an existing target rather than recomputing complete SIFT detection and matching. The earlier scripts repeatedly detect and describe features. The table cannot isolate an algorithmic speedup independent of this workload change, nor does it establish behaviour on a real day/night stream.

The bundled video provides a functional regression input. It tests decoding, feature processing, homography calls and output writing on the media supplied with the project. It lacks a ground-truth pose, framewise homography annotations and independent place labels. The number of nonnull estimates or warps therefore describes execution, not accuracy. Later benchmark experiments supply the missing ground-truth comparison where available.

3.6 Failure mechanisms motivating the redesign

The first limitation is a fixed reference identity. Every successful match is interpreted relative to one target image. There is no mechanism to select among stored places or attach multiple users' posts to a shared identity. Adding a gallery without explicit rejection would introduce false-place overlays, because even a nonmatching frame can have nearest neighbours.

The second limitation is that temporal continuity is used to bridge a recognition failure. Reuse and tracking help when the correct target has already been found, but they do not solve re-entry under a different appearance. A visitor arriving at night needs a fresh correspondence to the stored representation. That task motivates the learned local and global components evaluated here.

The third limitation is implicit state. Input configuration, recognition thresholds, cached geometry and rendering all live in script-level variables. A failure in one stage can be difficult to attribute. Separating extraction, matching, verification and compositing makes it possible to store the evidence supporting an anchor and to distinguish a wrong retrieval from a bad homography.

The fourth limitation is the absence of experimental accounting. Without fixed query lists, ground truth, saved predictions and runtime definitions, a demonstration can support only a narrow claim that an overlay appeared on selected footage. The redesign adds this accounting before drawing broader conclusions. Its aim is to establish which parts of the original idea are supported by measured evidence and which remain a research or deployment requirement.

Chapter 4

System Design and Architecture

4.1 End-to-end responsibilities

The implemented system divides a query into capture or decoding, preprocessing, place hypothesis generation, local correspondence, geometric verification, content lookup and rendering. The division follows the different evidence required at each stage. Capture supplies pixels; retrieval supplies candidate identities; geometry supplies a mapping; the database supplies content associated with the verified identity. A later stage is not allowed to turn an absent earlier result into an apparently successful anchor.

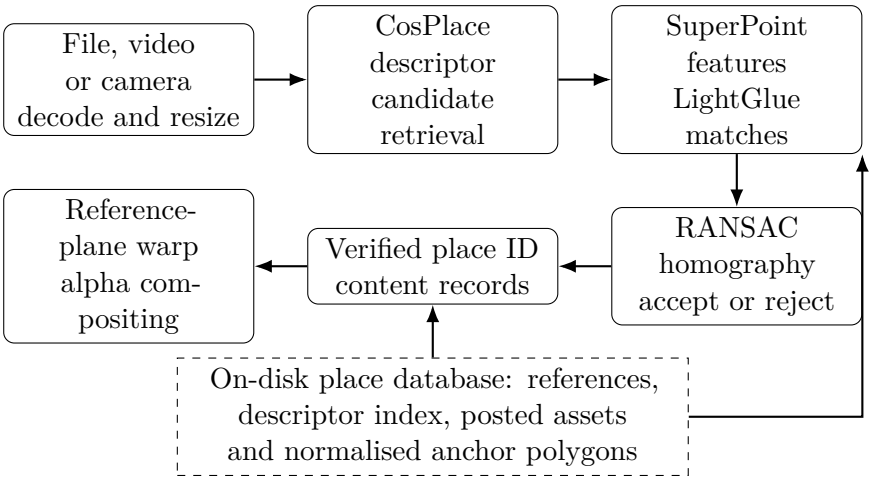


Figure 4.1: Implemented hierarchical recognition and planar content-rendering flow. The diagram is an architectural specification, not a measured throughput result.

Figure 4.1 depicts the implemented path. The classical mode replaces the global shortlist and learned correspondence stage with exhaustive SIFT–FLANN verification. The learned-local mode removes global retrieval but retains exhaustive SuperPoint–LightGlue comparisons. Keeping those alternatives behind the same query interface makes stage removal explicit rather than changing the application data model for each method.

4.2 Module contracts

4.2.1 Image I/O and coordinates

The I/O module returns a decoded BGR image and the transformation from original image coordinates to the working resolution. This second output is essential for evaluation: ground-truth homographies are expressed in the original images, whereas inference uses resized images. The module preserves aspect ratio and records separate horizontal and vertical scale values after integer rounding. It rejects unreadable files instead of returning an empty array that fails later in feature extraction.

The video iterator accepts a file path, numeric webcam index or stream URL. It stops on EOF or a failed read and releases the capture resource. An explicit frame limit supports bounded experiments. A production live-stream client could add a retry state with a deadline; that policy should be kept outside geometry and should not silently reuse the last camera frame as a new observation.

4.2.2 Local features and correspondences

A local-feature record contains pixel coordinates, descriptor arrays and any model-specific tensors required by the matcher. The shared interface returns two coordinate arrays and the descriptor-index pairs that produced them. Empty output has a stable shape. This allows geometry to reject insufficient support without depending on the internal format of either OpenCV or LightGlue.

Learned features are associated with the resized image, and internal extractor resizing is disabled for the controlled comparison. Reference features are cached when the pipeline is constructed. Query features are extracted once and reused across candidates. This division avoids measuring repeated extraction merely because a shortlist contains several references.

4.2.3 Geometry and verification

The geometry module returns the estimated homography, an inlier mask, count and ratio, and a separate acceptance flag. Retaining a rejected matrix for diagnostics is useful, but rendering checks the acceptance decision through the pipeline's selected prediction. A matrix existing in memory does not itself authorise an overlay. The interface also exposes corner projection and camera-pose error utilities for tests and future calibrated evaluations.

The acceptance policy combines sufficient correspondence support with numerical and boundary checks. This policy is common to the normalised local-method comparison. It is intentionally distinct from the original scripts' hard-coded thresholds, whose unmodified results are reported separately. The distinction prevents a software refactor from being mistaken for a perfectly controlled reproduction of every upstream heuristic.

4.2.4 Global retrieval

The global module loads the official CosPlace weights into the compatible ResNet18 and pooling architecture. Random ImageNet initialisation weights are not downloaded because the complete pretrained state is supplied by the selected checkpoint. A fixed preprocessing path resizes to the recorded descriptor input shape and applies channel normalisation. Gallery and query descriptors follow the same path.

Ranking uses the dot product of normalised vectors with stable tie ordering. Exact search is appropriate for the small galleries in this artifact and makes the evaluation easy to inspect. An approximate index would introduce another source of candidate recall loss; it should be evaluated separately before a larger deployment. The descriptor index stores image fingerprints so stale references cannot be silently matched against an obsolete cached vector.

4.3 Place-database schema

The database document has a schema version and a list of places. Each place has a unique identifier, one or more reference-image paths, and a list of posted-content records. Paths resolve relative to the database document so the artifact can be moved as a directory. The schema keeps a place identity distinct from a particular reference view: adding a second photograph need not duplicate all associated posts.

Listing 4.1: Implemented place-record structure with illustrative identifiers. The numerical anchor coordinates are configuration values, not experimental results.

```
{
  "schema_version": 1,
  "places": [{
    "id": "place_identifier",
    "references": ["reference_view.jpg"],
    "content": [{
      "id": "post_identifier",
      "kind": "image",
      "asset": "posted_image.png",
      "author": "example-user",
      "created_at": "2026-11-05",
      "anchor_normalized": [
        [0.2, 0.2], [0.8, 0.2],
        [0.8, 0.65], [0.2, 0.65]
      ],
      "opacity": 0.8
    }]
  }]
}
```

Listing 4.1 shows the implemented fields. Normalised polygon coordinates preserve the intended reference region when an image is resized. They describe a quadrilateral in the reference image, not a metric plane pose or a three-dimensional volume. The asset can retain its own alpha channel. The current renderer consumes image assets; the schema's content-kind field provides a point for future video, note and mesh renderers without claiming they are implemented.

The index is stored in a companion compressed NumPy file. It contains descriptors, reference

paths, image hashes, the encoder checkpoint hash and preprocessing settings, including the image resize limit. Loading checks the index version, reference list, fingerprint count, configuration and finite descriptor matrix. A changed reference image, checkpoint or resize limit requires rebuilding. Enforcing cache-size limits remains a service implementation requirement.

Each content polygon is associated with one reference photograph. An optional `reference` field selects that photograph; when omitted, the first reference is used. A post is rendered only through its associated reference. Adding a second photograph does not transfer polygon coordinates between views; that requires a validated geometric relation.

4.4 Implemented query behaviour

A query extracts its local features once. In full mode it also extracts a global descriptor and ranks references, selecting the configured shortlist. Each candidate receives local matching and RANSAC verification. Accepted candidates are ranked by inlier count and then inlier ratio. If none is accepted, the returned place identifier is absent. Otherwise, the selected reference and the content records belonging to that reference's place are returned together.

Rendering loads each selected asset, maps its corners to its reference-anchor polygon, composes that map with the reference-to-query homography, and alpha-composites into the query frame. This implements a complete local recognition-to-content path. The content is not selected by manually choosing an overlay after observing a query result. It is retrieved through the matched place record.

Candidate diagnostics accompany the decision. They record place identifiers, support counts, ratios and acceptance flags. These diagnostics make it possible to explain why a familiar scene did not display a post: the correct reference may be outside the shortlist, or its local support may be inadequate. A user interface could convert those states into a prompt to move the camera or improve lighting. The research CLI reports the evidence directly.

4.5 Server and device allocation

The implemented artifact runs entirely on the host CPU. For a mobile deployment, global retrieval could remain on device for a geographically limited gallery, while a server supplies additional map candidates. Local verification could also run on device after downloading sparse reference features. The trade-off depends on model size, bandwidth, privacy requirements and the measured phone execution backend; those quantities cannot be decided from desktop throughput alone.

A viable interaction loop should separate occasional recognition from high-frequency tracking. The original `tep3.py` already illustrates this principle for a known target. Learned recognition could establish or refresh the map relation, after which the native AR tracking system updates the view. This architecture would not require executing the full hierarchical recogniser at the display refresh rate. Nevertheless, tracking drift and recovery latency would need their own measurements before the system could be described as persistent.

4.6 Testability and failure containment

The module boundaries support tests with known transforms and synthetic outliers, independently of network downloads. The database can be inspected without loading a neural model. Rendering can be tested with an opaque black asset and a transparent asset. I/O failures can be reproduced through bounded files. These checks protect coordinate and state conventions that otherwise produce convincing but incorrect overlays.

The main containment rule is rejection on insufficient evidence. A missing descriptor set, failed estimate, singular transform or absent verified candidate returns no anchor. Qualitative demonstrations retain these failures visibly. The resulting behaviour may display less content than a permissive prototype, but it provides an interpretable basis for studying whether a future recognition improvement actually increases correctly supported anchors.

Chapter 5

Datasets and Data Preparation

5.1 Selection principles and evidence levels

The repository contains demonstration media but no labelled research dataset. Data acquisition was therefore part of the experiment. Selection prioritised reachable public files, variation relevant to the application, ground truth appropriate to the measured task, and a subset size compatible with the available host. Two complementary public collections were selected: HPatches for planar geometric evaluation and Tokyo query photographs for real cross-time place recognition. A generated set adds controlled photometric perturbations.

These collections supply different evidence levels. A labelled place identity establishes whether a retrieved record refers to the intended location. A homography establishes a planar pixel mapping. A calibrated camera pose and three-dimensional map would establish metric localisation error. The first two are present in the selected data; the third is not. The report does not infer missing ground truth from visual plausibility, GPS labels or a fitted matrix.

Table 5.1: Downloaded and generated evaluation collections. Pair counts refer to this artifact, not full public datasets.

Collection	Images	Pairs / queries	Gallery
HPatches subset	144	120	24
Tokyo cross-time subset	96	96	24
Synthetic transformations	90	90	26

Table 5.1 summarises the actual evaluation collections. The downloaded and generated image counts come from the data and preparation manifests. The HPatches archive was downloaded in full from the author's reachable mirror, while only selected sequences were extracted for evaluation. Tokyo images were retrieved as selected ZIP members using byte ranges; their complete metadata records were retained to make the selection rule inspectable. Download failures and the bounded full-archive attempt are preserved in the command journal.

5.2 HPatches subset

5.2.1 Acquisition and deterministic selection

The original HPatches host failed DNS resolution during the recorded download check. The dataset author's Hugging Face mirror was reachable and supplied the full sequence archive. The manifest records its URL, file hash, extracted-image hashes and per-sequence file counts. This distinguishes a verified local copy from a statement that a URL happened to exist when the thesis was written.

The selected evaluation set consists of the first twelve sequence names in lexicographic order for each of the illumination and viewpoint prefixes. Every selected sequence contributes its reference image, all five targets and their corresponding homography files. Thus 144 images and 120 reference-target pairs are evaluated. The choice uses names and metadata before inference, rather than selecting pairs on which one method appears stronger.

The images are resized only when loaded by the common I/O module. Original homographies remain unmodified on disk. Their transformation into the working-image coordinate convention is computed during evaluation. This avoids creating a second ambiguous ground-truth directory whose matrices might no longer correspond to the original files.

5.2.2 Illumination and viewpoint protocols

The two prefix groups are evaluated separately because they probe different failure mechanisms. Illumination sequences change the appearance of a scene while retaining a related view. Viewpoint sequences change the projective relationship and visible detail. The official dataset description motivates this distinction [3, 17]. No attempt is made to relabel every illumination pair as a day/night pair: the prefix is the authoritative task label available for this experiment.

For matching, each target is paired with its own sequence reference. For the custom retrieval task, the gallery contains one reference from every selected sequence, and a target's correct identity is its sequence name. This provides distractors and a reproducible image-retrieval exercise. It does not add geographic coverage or convert HPatches into a city-scale VPR benchmark. Related targets from one sequence also remain statistically dependent, which motivates clustered uncertainty estimates.

5.2.3 Rights and source attribution

HPatches combines sequences with source-specific attribution described by its maintainers. The repository's software licence should not be assumed to grant a blanket commercial licence for all imagery. The artifact retains source links and uses the data as a research input. Any distribution of extracted photographs or a public product built from them requires review of the original source terms. This limitation is documented rather than replaced with an unsupported "open data" label.

5.3 Tokyo cross-time subset

5.3.1 Available archive and labels

The accessible Tokyo archive contains camera-phone query images and CSV metadata. The metadata identifies image filenames, geographic positions, viewing-direction labels and capture-condition labels. The original dataset was introduced for place recognition under time-of-day and viewpoint change [4]. The public query URL was confirmed through the dataset-formatting documentation [23].

A direct full-archive transfer exceeded the chosen bounded timeout and was not treated as a successful dataset download. Instead, the downloader read the ZIP central directory and retrieved required members with HTTP byte ranges. It checked the decompressed size and the archive's CRC for each member before writing it. SHA256 fingerprints were then recorded for future reproduction. Partial archive files are not used as valid complete archives.

All available CSV members were downloaded before subset selection. Records were grouped by their exact latitude-longitude site key. A complete selected site needed a front-facing daytime image, a left-facing daytime image, a front-facing night image and a front-facing sunset image. The first 24 complete site keys in lexicographic order were retained. That produces 96 image files with a transparent, metadata-only selection rule.

Figure 5.1 illustrates the resulting condition coverage. Visual inspection confirms that the retained night examples are real night photographs rather than images created by gamma correction. It also shows that a left-facing daytime observation can depict substantially different scene content from the front-facing gallery image. This makes the day-to-day split a viewpoint stress test, not necessarily an easier control than the aligned cross-time pairs.

5.3.2 Reference-to-query direction

Condition names throughout the artifact follow the arrow from reference to query. Day-to-night uses a daytime front reference and a night front query. Night-to-day reverses those roles. Day-to-sunset uses a daytime front reference and a sunset front query. Day-to-day uses a daytime front reference and a daytime left query. There is one query per selected site in each of these protocols.

The gallery has one reference per selected site. Place correctness uses the stored site identity, not a distance threshold inferred after observing predictions. The reverse protocol is useful because extraction and matching may not behave symmetrically when query and reference roles change. It nevertheless reuses the same physical sites, so the protocols are related observations rather than independent geographic test sets.

5.3.3 Difference from the official benchmark

The official Tokyo retrieval task uses a street-view image database. The executed protocol uses query photographs as a small cross-time reference gallery. This changes database scale, appearance distribution, viewpoint alignment and distractor difficulty. Results are therefore labelled "Tokyo

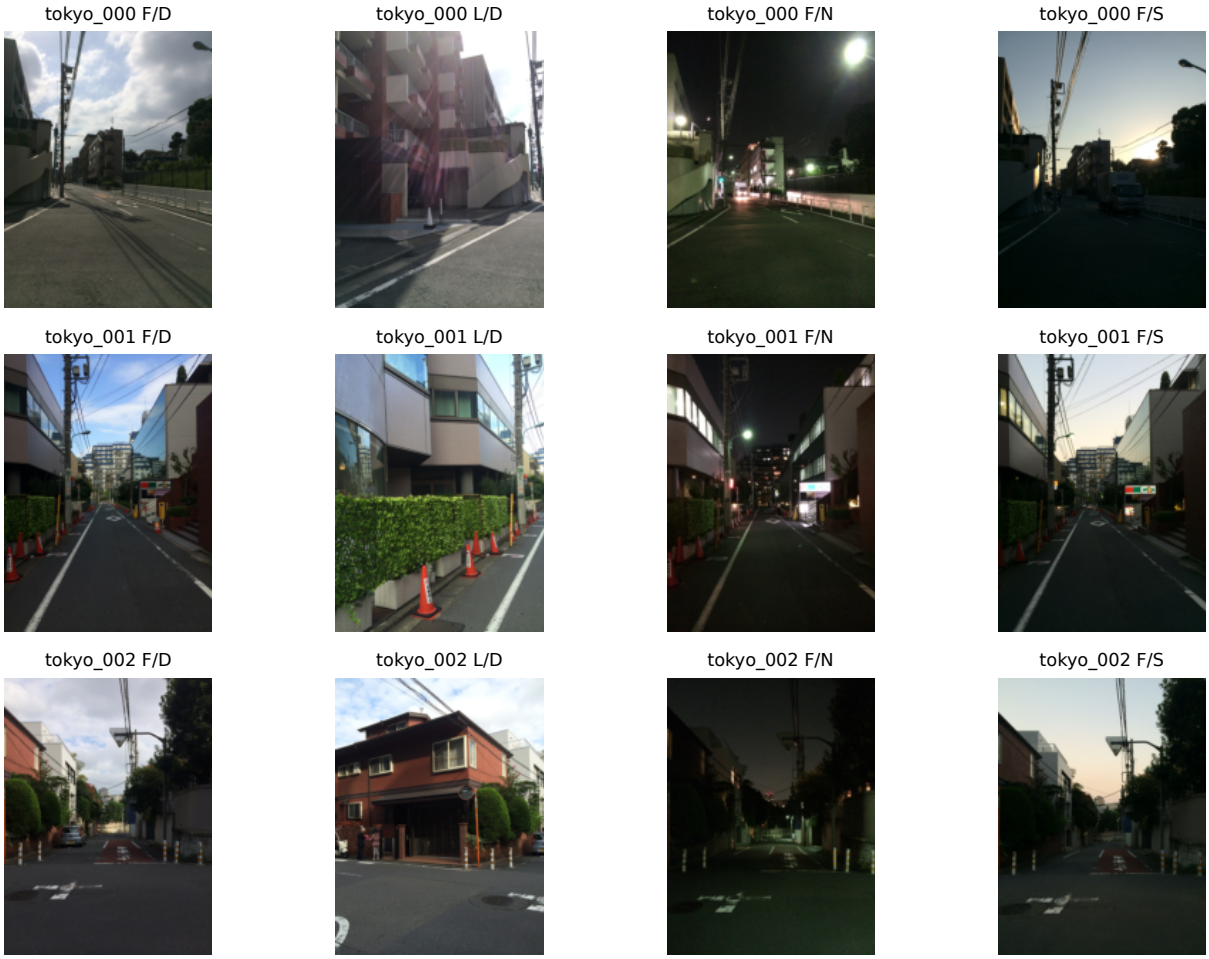


Figure 5.1: Actual downloaded photographs from the first selected Tokyo sites. Columns follow the retained front/day, left/day, front/night and front/sunset records. Orientation and condition labels come from the source metadata.

cross-time subset" throughout the thesis. They must not be compared directly with published official benchmark recall values.

The restriction is scientifically important when recall saturates. A small gallery with aligned front-facing views can be easy even though night recognition is difficult in a larger operational setting. Conversely, a deliberately different viewing direction can reduce overlap despite unchanged time of day. The experiment identifies these task properties rather than treating all appearance labels as a fixed difficulty ordering.

The query archive does not provide the calibrated camera poses or per-pair homographies needed for metric localisation or true planar-accuracy scoring. GPS site labels support recognition evaluation only. Geometric acceptance is reported, but a successful fitted homography is not assigned a metric translation error. The missing pose criteria are explicitly listed as unavailable in the results.

5.4 Synthetic appearance and camera perturbations

5.4.1 Source images and identity semantics

The synthetic set contains transformations of both repository images and selected HPatches references. Repository filenames do not establish independent real-world places. Their labels are therefore source-image identities. Retrieving a transformed `reference.jpg` means recognising its source image, not proving that the system has generalised to a new geographic location.

Six source images are transformed under five conditions and three fixed seeds, giving 90 generated queries. The gallery includes additional HPatches references as distractors. Every generated query has a known source identity and a known reference-to-query homography. The preparation manifest records the generated file hashes, seeds and benchmark-manifest hash so the exact test inputs can be regenerated and checked.

5.4.2 Photometric operations

The daytime condition changes brightness within a bounded range. Night conditions apply a gamma curve, reduced gain and additive noise. The severe condition uses a stronger attenuation; a warm-night condition additionally changes channel gains. The fog-like condition blends a bright low-contrast component and applies mild blur. These are explicit stressors, not calibrated physical camera or weather models.

Figure 5.2 provides an audit of their visual effect. The operations preserve scene geometry and do not introduce night-specific light sources, glare, moving objects or seasonal occlusion. Additive noise is not a full sensor noise model. A good result here establishes tolerance to these implemented perturbations only. Real Tokyo images provide a separate, narrower check under natural temporal appearance change.



Figure 5.2: Actual seeded synthetic queries generated from the two repository images. Columns show the evaluated photometric conditions. “Fog” denotes a contrast-and-blur simulation, not a real weather capture.

5.4.3 Known geometry and deterministic seeds

Before appearance modification, a small random projective warp perturbs the image corners. The seed determines both this warp and the subsequent photometric randomness. The homography is stored explicitly, including the scale mapping from the original reference to the working-resolution source. This detail prevents a synthetic experiment from accidentally comparing an estimate in resized coordinates with a ground-truth matrix in original coordinates.

The three query seeds support repeatability and sensitivity analysis. They do not create three independent physical scenes. Confidence statements must therefore preserve source identity as a grouping variable. Likewise, increasing the number of augmentations does not compensate for the small number of base photographs. The thesis reports those dependencies as validity constraints.

5.5 Training and evaluation separation

The optional adaptation experiment reserves the first four illumination and first four viewpoint sequences for training the projection head. Remaining selected sequences form the held-out HPatches retrieval evaluation. Training never uses Tokyo photographs. Additional synthetic training perturbations use a separate fixed seed list. The backbone remains frozen, and no test-based checkpoint selection is performed.

The main pretrained comparison may evaluate all selected HPatches sequences because none is used to train the supplied models within this artifact. The adaptation table uses only its held-out identities. Its numbers should therefore be compared with the pretrained descriptor reevaluated on that same held-out subset, not with a main-table average over a different set. The scripts

preserve both denominators to prevent that drift.

5.6 Open-set diagnostic

An additional diagnostic uses HPatches images as unknown queries against a Tokyo gallery. These images have no valid place identity in that gallery. A retrieval system still returns a ranking, but an anchoring system should reject all candidates unless there is sufficient evidence. The diagnostic reports acceptance as an error condition; retrieval recall is not meaningful when no correct gallery identity exists.

The collection-level difference makes this a limited open-set test. It does not represent a difficult nearby look-alike building, malicious image replay, or two nearly identical store fronts. Its purpose is to verify that the pipeline can return no anchor and to expose gross false acceptance. More realistic negative data are required before using the measured rejection behaviour as a safety assurance.

5.7 Integrity and idempotent acquisition

The downloader validates both archive-derived member properties and locally recorded hashes. Repeated execution checks existing files and skips a valid collection. Missing data can be restored and checked against the pinned manifest; an existing corrupt file produces an explicit error instead of silently changing the input. Archive paths are not extracted blindly: selected basenames and sequence directories determine destinations.

File-count checks enforce the expected image and homography structure. The final `benchmark.json` freezes protocol definitions before inference, and its hash is recorded with the run configuration. These controls do not prove that the original benchmark labels are error-free, but they establish exactly which input bytes and label rules produced the thesis results.

Chapter 6

Methodology

6.1 Notation and coordinate conventions

An original reference image is denoted I_r^o and an original query I_q^o. Their working-resolution forms are I_r and I_q. Pixel coordinates use the conventional horizontal-then-vertical ordering (x, y). A homogeneous point is $\tilde{\mathbf{x}} = (x, y, 1)^\top$, and the inhomogeneous projection is $\pi(a, b, c) = (a/c, b/c)$ when $c \neq 0$. Every estimated homography in the implementation maps reference pixels into query pixels.

Let S_r and S_q map original coordinates into their resized images. An original ground-truth map H^o is converted by

$$H_{r \to q} = S_q H^o S_r^{-1}. \tag{6.1}$$

This follows by substituting $\tilde{\mathbf{x}}_r^o = S_r^{-1}\tilde{\mathbf{x}}_r$ into the original mapping and then applying S_q. The implementation records actual rounded image dimensions, so S contains the realised horizontal and vertical ratios rather than a nominal scalar alone.

This convention also applies to generated queries. Their saved homography includes the original-reference-to-generation scale. Ground truth and estimated points are compared only after both have been expressed in the same working coordinate frame. A numerical unit test verifies a known projective transformation with outliers; the reproduction tests do not depend on benchmark-model downloads.

6.2 Method A: classical local matching

6.2.1 Feature extraction

For an image I, SIFT returns coordinates and descriptors

$$\mathcal{L}_A(I) = \{(\mathbf{x}_j, \mathbf{d}_j)\}_{j=1}^K, \qquad \mathbf{d}_j \in \mathbb{R}^{128}. \tag{6.2}$$

The representation is based on local gradient structure and scale/orientation normalisation [5]. The controlled experiment limits the requested feature budget through the common configuration.

The original script budgets and preprocessing remain in the separate upstream execution.

The normalised comparison converts each working image to grayscale and extracts features once. Empty descriptor sets are represented explicitly. No claim is made that the common budget yields an identical number of points for the two detectors: each detector selects its own repeatable structures, and SIFT may return multiple oriented features at related locations. The recorded feature and match counts describe the realised outputs.

6.2.2 Descriptor matching

For each reference descriptor $\mathbf{d}_i$, approximate search identifies its first and second query neighbours, $\mathbf{e}_{j_1}$ and $\mathbf{e}_{j_2}$. The tentative correspondence survives when

$$\|\mathbf{d}_i - \mathbf{e}_{j_1}\|_2 < \rho \, \|\mathbf{d}_i - \mathbf{e}_{j_2}\|_2 \,, \tag{6.3}$$

with ρ fixed in the configuration. FLANN settings match the recorded index strategy [6]. The matcher guards against fewer than two returned neighbours. Its output is a coordinate correspondence set, not a place decision.

To extend the one-reference baseline to place recognition, Method A verifies every gallery reference. Candidates are ranked by RANSAC inlier support and then by inlier ratio, with stable order for remaining ties. The best accepted candidate supplies the anchor. A ranking is still retained when no candidate is accepted so retrieval quality and anchoring rejection can be analysed separately.

6.3 Method B: pretrained learned local matching

SuperPoint supplies keypoint positions, descriptors and the feature tensors required by LightGlue [7]. The pretrained networks are loaded in evaluation mode and executed without gradient recording. The input is the same working-resolution grayscale image used by the comparison; additional internal resizing is disabled. The model is not fine-tuned in the main A–D experiment.

LightGlue returns a partial set of descriptor-index correspondences. Its contextual interaction can be written schematically as

$$(\widehat{\mathcal{D}}_r, \widehat{\mathcal{D}}_q) = \Phi_\theta(\mathcal{D}_r, \mathcal{D}_q, \mathcal{X}_r, \mathcal{X}_q), \tag{6.4}$$

followed by learned assignment and filtering. This notation describes the role of the pretrained matcher rather than rederiving all of its layers. The official model supplies the adaptive depth and feature-pruning behaviour [8]. Recorded configuration values identify the executed path.

Method B is evaluated in two ways. Oracle-pair evaluation supplies the correct reference pair so local correspondence quality can be isolated. Exhaustive gallery evaluation performs learned matching against every reference, allowing retrieval and verification to be compared with Method A on identical queries. Exhaustive matching is an informative accuracy reference but may have unacceptable query cost as the gallery grows.

6.4 Method C: global retrieval

6.4.1 Descriptor construction

The global encoder is the released CosPlace ResNet18 model with generalised mean pooling, a learned projection and final normalisation [11, 12]. If $F_{c,u,v}$ denotes a channel activation, the pooling operation is

$$g_c = \left(\frac{1}{UV} \sum_{u,v} \max(F_{c,u,v}, \epsilon)^p \right)^{1/p}. \tag{6.5}$$

The pretrained parameter p is loaded with the checkpoint. The global descriptor is

$$\mathbf{z}(I) = \frac{W\mathbf{g} + \mathbf{b}}{\|W\mathbf{g} + \mathbf{b}\|_2}, \qquad \mathbf{z} \in \mathbb{R}^{512}, \tag{6.6}$$

with the implementation retaining the official architecture's normalisation placement. Input channel means and scales are fixed, and gallery and query images use the same descriptor input size.

6.4.2 Ranking and identity

For a gallery matrix Z whose rows are normalised descriptors, similarity to a query is

$$s_i = \mathbf{z}(I_i)^\top \mathbf{z}(I_q). \tag{6.7}$$

Sorting these scores yields a complete reference ranking. A reference's place identifier is then used for correctness. With multiple references per place, a production evaluator would need an explicit reference-to-place aggregation policy; the implemented database supports multiple references, while the frozen benchmark galleries use one per identity.

Method C alone does not estimate a homography or render an anchored post. Its anchor-accuracy columns are consequently unavailable rather than zero-performance geometric estimates. The no-geometry ablation additionally treats its top retrieval as accepted, solely to expose the consequence of replacing verification with similarity. That variant does not produce a valid projective transform.

6.5 Robust homography estimation

6.5.1 Linear constraints and minimal evidence

A planar correspondence obeys

$$\widetilde{\mathbf{x}}_q \sim H\widetilde{\mathbf{x}}_r, \qquad H \in \mathbb{R}^{3\times 3}, \quad \operatorname{rank}(H) = 3. \tag{6.8}$$

The equivalence is up to a nonzero scalar. Expanding the cross-product constraint

$$\widetilde{\mathbf{x}}_q \times H\widetilde{\mathbf{x}}_r = \mathbf{0} \tag{6.9}$$

gives two independent linear constraints per point pair. A nondegenerate minimal sample has four correspondences. Collinear or nearly collinear evidence cannot constrain a stable general planar map. These projective principles motivate the rank and boundary checks used after robust fitting [13].

The comparison uses OpenCV's RANSAC homography estimation with a fixed pixel threshold, confidence and iteration cap. For a candidate H, forward transfer error is

$$e_j = \|\mathbf{x}_{q,j} - \pi(H\widetilde{\mathbf{x}}_{r,j})\|_2 \,. \tag{6.10}$$

The inlier mask indicates correspondences compatible with the fitted model at the configured threshold. The thesis retains this support even if the final acceptance policy rejects the matrix.

6.5.2 Sampling interpretation

For an idealised inlier fraction w, minimal sample size s and desired sampling confidence c, the familiar RANSAC relation is

$$N \geq \frac{\log(1-c)}{\log(1-w^s)}. \tag{6.11}$$

It follows by requiring the probability of never selecting an all-inlier sample, $(1-w^s)^N$, to be no larger than $1-c$ [15]. The relation illustrates why descriptor ambiguity can raise geometric-estimation cost. It is a theoretical interpretation, not a claim that each recorded call used exactly this number of iterations.

6.5.3 Acceptance policy

The verifier requires a minimum inlier count, a minimum inlier ratio, a finite full-rank map and an admissible projected boundary. After normalising the homogeneous scale, it rejects severe numerical conditioning, collapsed projected area and a reference boundary crossing the projective horizon. The latter is checked through denominator signs at the corners. Convexity of the projected quadrilateral supplies an additional sanity check.

Homogeneous scale normalisation precedes numerical threshold checks: multiplying a valid map by a small nonzero constant must not change its validity, projected points or rendered content. Anchor polygons are independently checked for finite, ordered, convex corners. Stored normalised coordinates must lie within the reference image; crossed or collapsed polygons are rejected before warping.

These conditions reduce obvious geometric failures but do not certify a correct physical anchor. A wrong repeated facade can produce a consistent planar map. A correct place can contain several depth layers that violate a single homography. A large region can project outside the visible query. The acceptance flag is therefore treated as a decision made by the prototype, while true homography success requires independent ground truth.

Algorithm 1 applies unchanged to the two normalised local methods. This is important for attributing an improvement to correspondence rather than to a weaker geometric acceptance rule.

Algorithm 1 Shared local verification

Require: Reference/query features, reference image shape, fixed thresholds
Match descriptors and recover coordinate pairs $(\mathbf{x}_{r,j}, \mathbf{x}_{q,j})$
if fewer than four pairs exist **then**
    return rejection with empty support
end if
Estimate H and an inlier mask with RANSAC
Count inliers and compute their fraction of tentative matches
Normalise H and test finiteness, rank, conditioning and boundary validity
return H, mask, support statistics, and the combined acceptance decision

The implementation stores failures as rows, so a method cannot improve its reported accuracy by excluding difficult pairs from the denominator.

6.6 Method D: hierarchical recognition and anchoring

The full pipeline ranks the gallery globally, selects a fixed-size shortlist, and performs Method B's local verification on that shortlist. It selects the accepted candidate with the strongest inlier support, breaking ties by ratio. An absent accepted candidate yields no place and no rendered post. The primary configuration uses the recorded shortlist size; additional runs vary that size and remove stages.

Algorithm 2 Hierarchical place and content query

Require: Query I_q, cached reference features, descriptor matrix Z, place database
Compute global query descriptor $\mathbf{z}_q$ and a ranked candidate list
Compute local query features once
for each of the first k reference candidates **do**
    Run Algorithm 1 against the cached reference features
    Retain diagnostics and any accepted transformation
end for
if no candidate is accepted **then**
    return no anchor and the candidate diagnostics
end if
Select the strongest accepted reference and its place identifier
Retrieve that place's posted-content records
Compose content-to-reference and reference-to-query transformations
Render content with explicit alpha and return the verified place

Algorithm 2 implements the hierarchy reviewed in Chapter 2. Global retrieval limits search, while geometry supplies the map needed by the renderer. It does not provide an independent world-scale estimate. Reference caching and one-time query extraction reduce duplicated work but do not remove the cost of matching several candidates.

6.6.1 A decomposition of failure

Let G denote that a usable correct reference enters the shortlist, L that sufficient correct local evidence is recovered, and V that the verifier accepts an adequate map. Then the desired event is bounded by the upstream stages:

$$P(G \cap L \cap V) = P(G)P(L \mid G)P(V \mid G, L). \tag{6.12}$$

This is a probability identity, not an independence assumption. It explains why increasing local accuracy cannot recover a reference excluded by retrieval, and why perfect retrieval cannot ensure anchoring when the images do not overlap. The ablation tables expose these failure pathways using recorded intermediate decisions.

False acceptance is a different event: a candidate may pass V without being the correct place or without satisfying ground-truth planar accuracy. Reporting only the acceptance rate would reward such failures. The results therefore pair acceptance with correctly verified identity and, where available, actual corner-error success.

6.7 Content coordinates and rendering

A content asset has its own pixel domain. Let $A_{c\to r}$ map its four corners to a normalised reference-anchor quadrilateral converted into reference pixels. The composed transform is

$$H_{c\to q} = H_{r\to q}A_{c\to r}. \tag{6.13}$$

This order follows directly from applying the content-to-reference mapping first. A reversed order would generally produce an unrelated projection and is not interchangeable even though both matrices are 3×3.

For premultiplied content colour αC and warped alpha α', compositing is

$$I_{\text{out}} = \text{warp}(\alpha C, H_{c\to q}) + (1 - \alpha')I_q. \tag{6.14}$$

An asset without an alpha channel receives an explicit opaque mask before the configured opacity is applied. This preserves black pixels as valid content. Rendering is skipped when the selected transformation is invalid or no candidate is accepted. The current application supports planar image posts; world-space occlusion and illumination-consistent three-dimensional rendering are future requirements.

6.8 Short appearance-adaptation experiment

6.8.1 Restricted learning scope

The optional experiment freezes the CosPlace backbone and learns a bias-free projection $W_h \in \mathbb{R}^{128\times512}$. The output is

$$\mathbf{u}(I) = \frac{W_h\mathbf{z}(I)}{\|W_h\mathbf{z}(I)\|_2}. \tag{6.15}$$

This is both descriptor compression and adaptation. Its comparison cannot isolate those effects completely, so the report avoids describing it as full-network day/night fine-tuning. The training-place list, fixed seeds and final head checkpoints are retained.

6.8.2 Triplet loss and sampling

For anchor, positive and negative descriptors from the selected training places, the implemented loss is

$$\mathcal{L} = \frac{1}{B}\sum_{b=1}^{B}\max(0, d(\mathbf{u}_a,\mathbf{u}_p) - d(\mathbf{u}_a,\mathbf{u}_n) + m), \qquad d(\mathbf{a},\mathbf{b}) = 1 - \mathbf{a}^\top\mathbf{b}. \tag{6.16}$$

The positive shares the source-place label and excludes the same sample index; the negative comes from another training place. The margin, optimiser, step count and learning rate are fixed in the script. One condition uses available real sequence images only; the other adds synthetic night variants. No held-out score selects a checkpoint or adjusts the step count.

Algorithm 3 Frozen-descriptor projection training

Require: Disjoint training/test places, fixed seed, frozen encoder, step budget
1: Extract training descriptors and labels; optionally add seeded night variants
2: Initialise a 512-to-128 linear head from the specified seed
3: **for** each optimiser step **do**
4: Sample anchors, different positive samples from the same place, and negative places
5: Normalise projected descriptors and compute the triplet loss
6: Update only the projection head
7: **end for**
8: Save the final head and loss trace
9: Evaluate held-out HPatches places and all selected Tokyo protocols once

Algorithm 3 is repeated for three fixed initialisations. Reporting the mean, variability and small-sample interval prevents selecting a favourable seed. A low training loss is not treated as evidence of better place recognition outside the training domain. The chosen experiment is deliberately short and interpretable; its limitations are part of the result.

Chapter 7

Experimental Setup

7.1 Recorded execution environment

The experiments ran on the available Apple M2 laptop with the memory configuration recorded in `results/environment.json`. The software environment is a clean project-local CPython virtual environment. The initial default interpreter was unsuitable for the pinned dependency set, so the available system CPython interpreter was used. Dependency versions were installed and recorded rather than assumed from pre-existing user packages.

The runtime reported neither CUDA nor an available MPS backend for these runs. All reported inference therefore uses the CPU. Both PyTorch and OpenCV are configured to use two threads in the controlled scripts. This restriction makes the workload explicit and avoids silently changing thread counts between methods. It does not establish that this is the fastest possible host configuration.

The primary installed versions are listed in the environment artifact and dependency specification. The reproducibility bundle also records the official LightGlue and CosPlace code revisions and SHA256 hashes of all downloaded checkpoints. Model parameter counts and checkpoint sizes are obtained from actual loaded models and files. They are not inferred by counting the number of layers in a paper diagram.

Table 7.1: Measured parameter counts and downloaded checkpoint sizes. Checkpoints include serialisation metadata.

Model	Parameters	Checkpoint MB
CosPlace	11439169	45.83
SuperPoint	1300865	5.21
LightGlue	11851601	47.50

Table 7.1 separates trainable-parameter count from serialised checkpoint size. Checkpoint bytes include format metadata and any stored buffers. Runtime memory is larger because it includes model objects, image tensors, local features, library imports and allocator state. A separate complete-process peak measurement is consequently reported in the deployment analysis.

7.2 Configuration and fixed controls

Table 7.2: Executed comparison configuration. Values are explicit experimental settings, not measured outcomes.

Setting	Value
Working-image maximum side	640 pixels
Requested local-feature budget	512
SIFT ratio threshold	0.70
RANSAC transfer threshold	5 working pixels
RANSAC confidence / maximum iterations	0.999 / 3000
Minimum inliers / inlier ratio	12 / 0.20
Primary full-pipeline shortlist	3 references
Shortlist ablations	1 and 5 references
CosPlace input / descriptor	224×224 / 512 dimensions
PyTorch / OpenCV threads	2 / 2
Main seed	17
Synthetic-query and training-head seeds	17, 23, 42

Table 7.2 states the settings shared across compared methods. The feature budget and RANSAC thresholds are not tuned separately on each test split. Reference extraction is cached. Query local features are extracted once per method and reused across the gallery. Global-only retrieval follows its own smaller input shape, which is documented because global descriptors and pixel correspondences have different computational needs.

PyTorch, NumPy and OpenCV random states are set. Candidate matching receives a deterministic seed derived from its method and image-path identities so repeated comparisons do not depend on the order in which a cache is populated. This improves repeatability within the installed environment. Floating-point kernels and library changes can still produce differences on another machine, so exact input hashes and environment versions remain necessary.

7.3 Evaluation populations

The frozen benchmark contains 306 labelled positive reference-query pairs, which produce 306 measurements per local method and therefore twice that many oracle-pair rows. The query protocols additionally include unknown-place diagnostics. The executed exhaustive comparison produced 2226 query-method prediction rows. These counts are generated from the saved preparation and result files.

Table 7.3: Frozen evaluation protocols. Reference-to-query direction is used in condition names.

Protocol	Gallery	Queries
hpatches_illumination	24	60
hpatches_viewpoint	24	60
tokyo_day_to_day	24	24

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Table 7.3 (continued)

Protocol	Gallery	Queries
tokyo_day_to_night	24	24
tokyo_night_to_day	24	24
tokyo_day_to_sunset	24	24
synthetic_day	26	18
synthetic_night	26	18
synthetic_night_severe	26	18
synthetic_warm_night	26	18
synthetic_fog	26	18
open_set	24	12

Table 7.3 gives the denominator for each protocol. The same query list is evaluated by A, B, C and the full pipeline. Stage ablations use those same lists. A query failure remains in its denominator; it is not removed because a model produced no descriptors or a homography could not be estimated.

The principal unit of evidence is a place or source image rather than an individual pixel correspondence. HPatches targets from one sequence share a reference and scene. Synthetic variants share a source image. Tokyo condition splits reuse physical sites. These dependencies limit the interpretation of aggregate counts, especially when uncertainty intervals are calculated from a small, selected set.

7.4 Retrieval metrics

Let Q be a query set and $\mathcal{P}(q)$ the valid gallery-place identities for query q. With ranked identities $r_1(q), r_2(q), \ldots$, recall is

$$R@k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\{\mathcal{P}(q) \cap \{r_1(q), \ldots, r_k(q)\} \neq \varnothing\}. \tag{7.1}$$

The implemented benchmark has one correct place identity per positive query. Recall@1, @5 and @10 share a query denominator and use the same ordered result list. Larger values of k are necessarily easier in these small galleries, so high Recall@10 is not interpreted as high-precision anchoring.

For exhaustive local methods, the ranking uses inlier support even if the top candidate is not accepted. This permits a correct retrieval with insufficient evidence to render. For the full pipeline, shortlisted references are reranked locally and the remaining global ranking is retained as an unverified tail. Its Recall@5 or @10 may therefore include references that were never locally verified. Correctly verified-place rate is reported separately to show actual decisions.

The global-only method does not declare an anchor. It is inappropriate to compare its absence of a geometric result with a zero homography success rate as if a model had attempted and failed to fit a transform. Tables mark that field unavailable. The no-geometry ablation instead

explicitly makes an unverified acceptance decision, allowing its false-acceptance consequences to be measured.

7.5 Matching and planar geometry metrics

7.5.1 Correspondence support

For M tentative matches and N RANSAC inliers, inlier ratio is N/M when $M > 0$ and zero otherwise. Median inlier counts and mean ratios are computed across all evaluated positive pairs, including failures. A high ratio from a very small correspondence set is not equivalent to a well-supported surface map, which is why the verifier also requires an absolute count.

Geometric acceptance rate is the fraction of pairs satisfying the configured support and validity conditions. On Tokyo it is the strongest available geometric decision metric, but it is not ground-truth alignment accuracy. On HPatches and synthetic pairs, independent ground-truth homographies make a stronger evaluation possible.

7.5.2 Corner-error success

Let $\mathbf{c}_1, \ldots, \mathbf{c}_4$ be the working reference corners. Mean corner error is

$$e_H = \frac{1}{4} \sum_{i=1}^{4} \left\| \pi(\widehat{H}\widetilde{\mathbf{c}}_i) - \pi(H^{\mathrm{gt}}\widetilde{\mathbf{c}}_i) \right\|_2 . \tag{7.2}$$

An oracle-pair homography success requires acceptance and $e_H \leq 5$ working pixels. Invalid estimates are failures. For cumulative error curves, they remain at infinite error so the curve denominator includes all queries. A conditional median over finite errors is interpreted only alongside the failure rate.

End-to-end planar success additionally requires the selected place to be correct. A highly accurate homography to a wrong reference is not counted. This stricter metric measures the chain from candidate selection to anchoring, while oracle-pair success isolates local geometry when the correct reference is supplied. Their difference reveals search-stage losses.

7.6 Timing, memory and model-size protocols

The upstream measurements time bounded script execution including initialisation and file I/O. The main evaluation records feature-extraction and matching-plus-geometry times, reusing cached reference computation. Its query stage sum adds the realised query extraction and selected pair costs; it is useful for accounting but excludes some application work and is not a direct end-to-end stopwatch measurement.

The separate runtime profiler starts an isolated process for each method. It builds the reference cache, warms the path, and times actual query and rendering calls on a fixed small set of pre-decoded Tokyo images with repeated execution. It reports median, mean and upper-tail latency,

and stores every timing sample. Asset loading during rendering remains inside the measured call. Camera decoding, network transfer and display are excluded.

Peak RSS is obtained from the operating system's process resource accounting, with platform-specific units converted into bytes. It includes imports, model weights, gallery caches and transient allocation. It is not presented as the exclusive tensor memory of the selected model. Serialised model size is measured from file length and reported separately. No phone power, thermal, battery or neural-accelerator measurement is claimed.

7.7 Uncertainty and statistical interpretation

For a binary site-level outcome with k successes among n sites, the report computes a Wilson interval. With $\widehat{p} = k/n$ and a normal quantile z, its centre and half-width are

$$\mu_W = \frac{\widehat{p} + z^2/(2n)}{1 + z^2/n}, \tag{7.3}$$

$$h_W = \frac{z}{1 + z^2/n}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n} + \frac{z^2}{4n^2}}. \tag{7.4}$$

The interval remains nonzero even when every selected query succeeds. It describes finite-sample uncertainty under an independent-site approximation, not a correction for the metadata-based geographic selection or the changed Tokyo benchmark task.

HPatches uncertainty is also computed with a cluster bootstrap that samples sequence identities and includes their associated targets. Synthetic variants are discussed as repeated observations of a few sources, not independent places. For projection training, three fixed seeds yield a mean, sample standard deviation and a small-sample t interval. With so few seeds the interval is unstable; it communicates uncertainty rather than establishing a precise population effect.

No significance claim is based on selecting the best seed, dropping rejected queries or treating thousands of pixel matches as independent successful places. No hyperparameters are chosen using the test-set result tables. The design is an exploratory reproducible comparison, with the limited data coverage made explicit.

7.8 Ablations and failure inspection

The shortlist-size ablation compares one, three and five candidates. Exhaustive learned matching removes the global stage, and global-only retrieval removes local geometric evidence. A no-geometry acceptance diagnostic shows the distinction between a candidate rank and a supported anchor. Projection training compares available sequence images with and without extra synthetic night variants. Export validation compares native, float ONNX and QDQ INT8 descriptors.

Qualitative selections follow documented rules: prefer a case where learned local verification succeeds while SIFT rejects, otherwise use a fixed fallback. Correspondence coordinates and masks are saved by an actual rerun, and overlay figures use saved estimates. This is a deliberately

illustrative selection rather than a random sample of success. Failure examples and aggregate tables prevent the selected demo from standing in for overall accuracy.

Chapter 8

Results and Discussion

8.1 Reading the evidence

The results are derived from the saved predictions, pair statistics, training records, export outputs and runtime profiles. All empirical tables and plots in this chapter are generated by `scripts/generate_artifacts.py`. The generation step records hashes of its input result files. Headline values in the prose use generated LaTeX macros so a rerun cannot silently leave those values inconsistent with the tables.

Three outcomes are kept separate throughout. Retrieval asks whether the correct identity appears in a ranked list. Correct verification asks whether the pipeline accepts the correct identity. Planar success additionally asks whether the accepted transformation agrees with an independent ground-truth homography. A result may satisfy one outcome without satisfying the next. This distinction is especially important for Tokyo, where no true planar or metric-pose error can be computed from the downloaded metadata.

8.2 Oracle-pair local matching

Table 8.1: Oracle-pair matching under shared geometry settings. Inliers are medians, ratios are means over all pairs; H_5 requires accepted geometry and mean corner error at most five resized pixels. Tokyo has no homography ground truth.

Protocol	Method	n	Inliers	Ratio (%)	Accept (%)	H_5 (%)
HP illum.	SIFT	60	58.5	83.7	83.3	81.7
HP illum.	SP+LG	60	317.0	94.6	100.0	100.0
HP view.	SIFT	60	97.5	89.8	93.3	86.7
HP view.	SP+LG	60	218.0	88.9	90.0	86.7
Tokyo D–D	SIFT	24	9.0	51.7	41.7	—
Tokyo D–D	SP+LG	24	39.0	61.1	54.2	—
Tokyo D–N	SIFT	24	8.5	64.3	33.3	—

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Table 8.1 (continued)

Protocol	Method	n	Inliers	Ratio (%)	Accept (%)	H_5 (%)
Tokyo D–N	SP+LG	24	169.0	84.7	100.0	—
Tokyo N–D	SIFT	24	7.0	64.9	33.3	—
Tokyo N–D	SP+LG	24	169.5	86.0	100.0	—
Tokyo D–S	SIFT	24	35.0	73.7	66.7	—
Tokyo D–S	SP+LG	24	204.0	84.2	100.0	—
Synth. day	SIFT	18	255.0	95.5	100.0	100.0
Synth. day	SP+LG	18	416.5	96.5	100.0	100.0
Synth. night	SIFT	18	130.5	82.0	94.4	88.9
Synth. night	SP+LG	18	388.5	95.3	100.0	100.0
Synth. severe	SIFT	18	10.0	41.8	33.3	33.3
Synth. severe	SP+LG	18	188.0	87.6	100.0	100.0
Synth. warm	SIFT	18	101.0	79.5	88.9	88.9
Synth. warm	SP+LG	18	375.0	96.0	100.0	100.0
Synth. fog	SIFT	18	76.0	84.5	100.0	100.0
Synth. fog	SP+LG	18	402.0	96.3	100.0	100.0

Table 8.1 compares SIFT–FLANN and SuperPoint–LightGlue on identical positive image pairs with common geometry settings. Supplying the correct reference removes candidate-search failure from this comparison. The inlier statistics describe each method's own tentative matches, so a higher count is interpreted alongside acceptance and corner-error success rather than as a standalone proof of accuracy.

On the selected HPatches illumination pairs, SIFT achieves 81.7% accepted five-pixel homography success, while the learned combination achieves 100.0%. This supports an illumination-robustness benefit on this subset. The result remains conditional on the selected scenes, resolution and verification policy. It does not show that every illumination condition is solved or that the matcher can reconstruct detail absent from an underexposed image.

On the viewpoint pairs, the success values are 86.7% for SIFT and 86.7% for the learned combination. The learned method therefore does not dominate every geometric condition in the tested set. In particular, accepted estimates can still extrapolate poorly to reference corners outside the supported region. This observation cautions against using a greater number of matches as a substitute for evaluated alignment accuracy.

Figure 8.1 emphasises the different illumination and viewpoint outcomes. The illumination gain is consistent with the motivation for learned features, while the viewpoint result provides a negative control against an unconditional improvement claim. Different subsets or thresholds may change the comparison, and the artifact makes those choices explicit rather than treating its rates as universal model properties.

8.3 Place retrieval on real Tokyo photographs

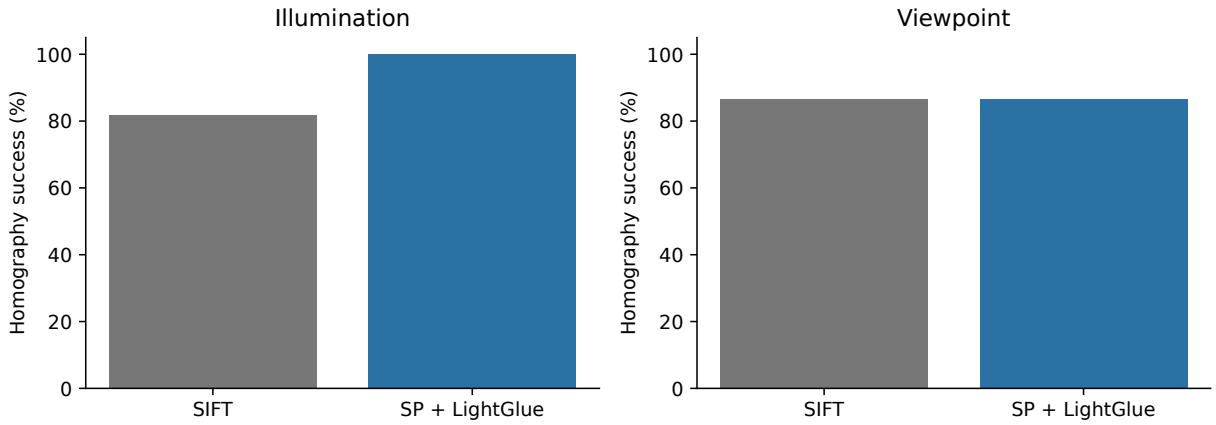


Figure 8.1: Measured oracle-pair homography success on the selected real HPatches sequences. A success requires accepted geometry and mean corner error no greater than five working pixels.

Table 8.2: Retrieval and correctly verified place rates (percent) on Tokyo protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
day_to_day	A: SIFT	54.2	66.7	75.0	41.7
day_to_day	B: SP+LG	58.3	70.8	75.0	54.2
day_to_day	C: CosPlace	41.7	62.5	75.0	—
day_to_day	D: Full ($k = 3$)	41.7	62.5	75.0	37.5
day_to_night	A: SIFT	58.3	91.7	91.7	33.3
day_to_night	B: SP+LG	100.0	100.0	100.0	100.0
day_to_night	C: CosPlace	100.0	100.0	100.0	—
day_to_night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night_to_day	A: SIFT	50.0	70.8	91.7	33.3
night_to_day	B: SP+LG	100.0	100.0	100.0	100.0
night_to_day	C: CosPlace	100.0	100.0	100.0	—
night_to_day	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
day_to_sunset	A: SIFT	83.3	95.8	100.0	66.7
day_to_sunset	B: SP+LG	100.0	100.0	100.0	100.0
day_to_sunset	C: CosPlace	100.0	100.0	100.0	—
day_to_sunset	D: Full ($k = 3$)	100.0	100.0	100.0	100.0

Table 8.2 gives the custom cross-time retrieval results. Day-to-night Recall@1 is 58.3% for exhaustive SIFT, 100.0% for exhaustive learned local matching, and 100.0% for CosPlace. The full hierarchy correctly verifies 100.0% of those selected queries. The corresponding SIFT correctly verified-place rate is 33.3%, which is lower than its retrieval recall because some correct ranked candidates lack enough accepted local evidence.

The saturation of learned and global recall on this split is a property of the selected protocol. It uses aligned front-facing observations and a gallery of 24 selected sites. It does not use the official

street-view database. Small-sample intervals and the task distinction are therefore necessary qualifications, not minor reporting details. Perfect observed recall in this setting cannot establish perfect recognition over a city or across arbitrary camera directions.

The daytime viewpoint-change result exposes a complementary weakness. CosPlace Recall@1 falls to 41.7%, and the primary full pipeline correctly verifies 37.5% of queries. Exhaustive SIFT correctly verifies 41.7%, and exhaustive learned matching verifies 54.2%. The hierarchy therefore sacrifices some recoverable cases when its shortlist excludes a useful reference. This is a measured negative result for the chosen global stage and shortlist policy.

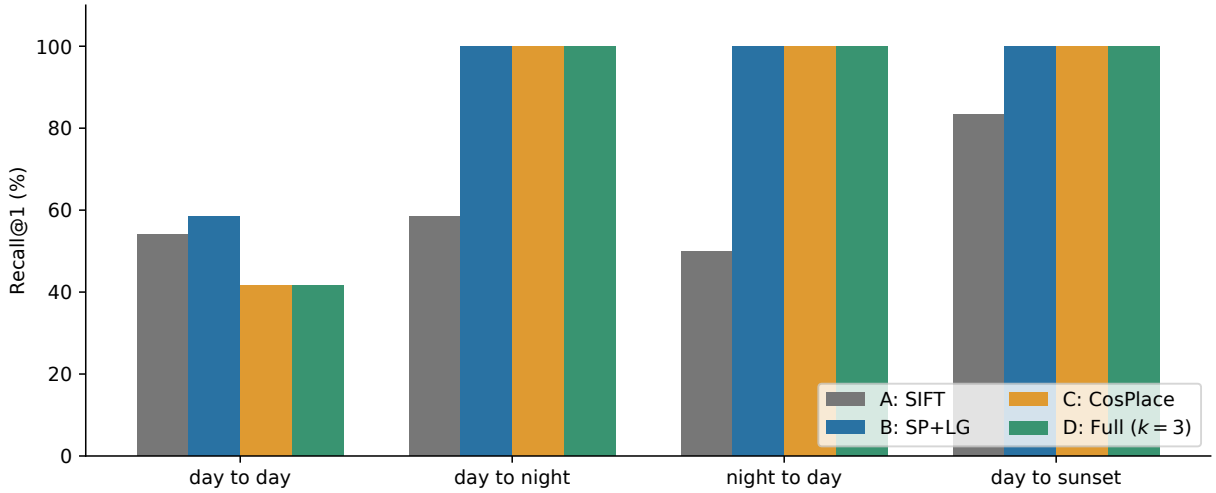


Figure 8.2: Recall@1 on the actual selected Tokyo query photographs. Day-to-day uses a different labelled viewing direction; the other splits use the retained front-facing records across time.

Figure 8.2 shows that the split label alone does not determine difficulty. Appearance change with similar viewpoint can be easier than unchanged daylight with reduced overlap. This supports retaining both kinds of variation in future validation. It also suggests that collecting several reference directions may be more useful than adding another photometric variant of one view, although that proposed collection strategy has not been tested here.

Night-to-day and sunset results are reported separately in the table because descriptor extraction and matching are direction-sensitive. They should not be counted as independent additional cities or sites. Their shared identities and reference observations limit any claim about geographic generalisation. The evidence supports behaviour under the recorded cross-time pairings.

8.4 HPatches retrieval and end-to-end planar success

Table 8.3: Retrieval and correctly verified place rates (percent) on HPatches protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
illumination	A: SIFT	90.0	98.3	98.3	81.7
illumination	B: SP+LG	98.3	100.0	100.0	98.3
illumination	C: CosPlace	100.0	100.0	100.0	—
illumination	D: Full ($k = 3$)	98.3	100.0	100.0	98.3
viewpoint	A: SIFT	96.7	98.3	100.0	93.3
viewpoint	B: SP+LG	91.7	93.3	93.3	90.0
viewpoint	C: CosPlace	71.7	98.3	100.0	—
viewpoint	D: Full ($k = 3$)	83.3	98.3	100.0	78.3

Table 8.3 evaluates retrieval among the selected HPatches reference images. Its scene-identity labels are sequence labels, not geographic locations. The task is useful for checking whether a coarse descriptor identifies a plausible reference before geometric verification, but it remains a custom distractor gallery rather than a standard place-recognition leaderboard.

Table 8.4: End-to-end planar success: the selected place must be correct, its geometry accepted, and mean corner error at most five working pixels. This differs from the oracle-pair matching table.

Protocol	Method	n	Correct place (%)	H_5 (%)
HP illum.	A: SIFT	60	81.7	80.0
HP illum.	B: SP+LG	60	98.3	98.3
HP illum.	D: Full ($k = 3$)	60	98.3	98.3
HP view.	A: SIFT	60	93.3	86.7
HP view.	B: SP+LG	60	90.0	86.7
HP view.	D: Full ($k = 3$)	60	78.3	78.3
Synth. day	A: SIFT	18	100.0	100.0
Synth. day	B: SP+LG	18	100.0	100.0
Synth. day	D: Full ($k = 3$)	18	100.0	100.0
Synth. night	A: SIFT	18	94.4	88.9
Synth. night	B: SP+LG	18	100.0	100.0
Synth. night	D: Full ($k = 3$)	18	100.0	100.0
Synth. severe	A: SIFT	18	33.3	33.3
Synth. severe	B: SP+LG	18	100.0	100.0
Synth. severe	D: Full ($k = 3$)	18	100.0	100.0
Synth. warm	A: SIFT	18	88.9	88.9
Synth. warm	B: SP+LG	18	100.0	100.0
Synth. warm	D: Full ($k = 3$)	18	100.0	100.0
Synth. fog	A: SIFT	18	100.0	100.0

Continued on next page

Table 8.4 (continued)

Protocol	Method	n	Correct place (%)	H_5 (%)
Synth. fog	B: SP+LG	18	100.0	100.0
Synth. fog	D: Full ($k = 3$)	18	100.0	100.0

Table 8.4 adds the complete planar-success criterion. A correctly selected identity may still fail the five-pixel condition. Likewise, good oracle-pair accuracy does not guarantee that the end-to-end system finds that reference. This table is the appropriate evidence when the claim concerns an accurately anchored planar post rather than recognition alone.

The comparison also illustrates a property of geometric verification: it can reject a correct place for application-valid reasons. A query may show a small region of the scene with insufficient spatial extent for stable extrapolation to the full reference plane. The appropriate response for the current renderer is to withhold the overlay, even though a broader semantic recogniser might confidently name the street. Future region-specific anchor evaluation could distinguish this case from complete recognition failure.

8.5 Synthetic appearance stress tests

Table 8.5: Retrieval and correctly verified place rates (percent) on synthetic protocols. Global-only retrieval does not produce an anchor. Full-pipeline recall uses its reranked complete list, with the unverified tail retained.

Split	Method	R@1	R@5	R@10	Verified
day	A: SIFT	100.0	100.0	100.0	100.0
day	B: SP+LG	100.0	100.0	100.0	100.0
day	C: CosPlace	100.0	100.0	100.0	—
day	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night	A: SIFT	100.0	100.0	100.0	94.4
night	B: SP+LG	100.0	100.0	100.0	100.0
night	C: CosPlace	100.0	100.0	100.0	—
night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
night_severe	A: SIFT	61.1	83.3	83.3	33.3
night_severe	B: SP+LG	100.0	100.0	100.0	100.0
night_severe	C: CosPlace	100.0	100.0	100.0	—
night_severe	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
warm_night	A: SIFT	88.9	100.0	100.0	88.9
warm_night	B: SP+LG	100.0	100.0	100.0	100.0
warm_night	C: CosPlace	100.0	100.0	100.0	—
warm_night	D: Full ($k = 3$)	100.0	100.0	100.0	100.0
fog	A: SIFT	100.0	100.0	100.0	100.0
fog	B: SP+LG	100.0	100.0	100.0	100.0

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Table 8.5 (continued)

Split	Method	R@1	R@5	R@10	Verified
fog	C: CosPlace	100.0	100.0	100.0	—
fog	D: Full ($k = 3$)	100.0	100.0	100.0	100.0

Table 8.5 reports the seeded transformations as their own collection. Learned methods retain strong source-image recognition under the implemented night stressors, while severe attenuation reduces SIFT's supported decisions. This is consistent with the implementation-level motivation, but the dependence on a small set of base images remains substantial.

Synthetic night retains the exact original scene content and applies a known warp. It cannot reproduce all changes in visibility, illumination direction, light-source placement or sensor response. The fog-like transformation likewise does not establish weather robustness. The actual input examples in Figure 5.2 define the scope of the claim more precisely than the condition names alone.

The three seeds vary random perturbations, not the physical scene population. Treating all generated frames as independent places would overstate evidence. For this reason, the report does not combine synthetic and real-night queries into one headline "night robustness" score. Such an aggregate would obscure both the shared-source dependence and the different ground-truth availability.

8.6 Stage ablations

Table 8.6: Stage ablations on identical queries. Exhaustive learned matching removes global retrieval; the no-geometry variant accepts the highest global result without an estimated transform.

Protocol	Variant	R@1	Correct (%)	Accept (%)
Tokyo D–D	B: SP+LG	58.3	54.2	75.0
Tokyo D–D	Full ($k = 1$)	41.7	37.5	41.7
Tokyo D–D	D: Full ($k = 3$)	41.7	37.5	50.0
Tokyo D–D	Full ($k = 5$)	41.7	41.7	54.2
Tokyo D–D	No geometry	41.7	41.7	100.0
Tokyo D–N	B: SP+LG	100.0	100.0	100.0
Tokyo D–N	Full ($k = 1$)	100.0	100.0	100.0
Tokyo D–N	D: Full ($k = 3$)	100.0	100.0	100.0
Tokyo D–N	Full ($k = 5$)	100.0	100.0	100.0
Tokyo D–N	No geometry	100.0	100.0	100.0
Synth. severe	B: SP+LG	100.0	100.0	100.0
Synth. severe	Full ($k = 1$)	100.0	100.0	100.0
Synth. severe	D: Full ($k = 3$)	100.0	100.0	100.0
Synth. severe	Full ($k = 5$)	100.0	100.0	100.0

Continued on next page

Table 8.6 (continued)

Protocol	Variant	R@1	Correct (%)	Accept (%)
Synth. severe	No geometry	100.0	100.0	100.0

Table 8.6 compares candidate limits and stage removal. Exhaustive learned matching provides the no-global-stage reference. Increasing the shortlist permits more comparisons but does not guarantee more correct accepted places: additional candidates can also create competing or incorrect geometric hypotheses. The measured viewpoint split shows that candidate recall and local support must be considered jointly.

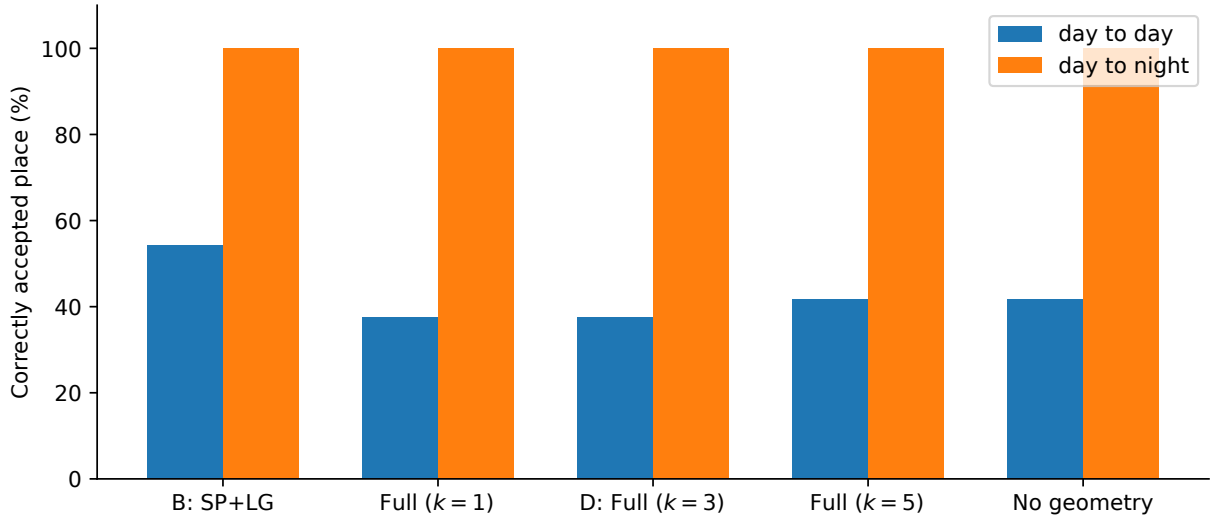


Figure 8.3: Measured correctly accepted-place rates for retrieval and geometry ablations. "No geometry" is a diagnostic acceptance policy without an anchored transform.

Figure 8.3 makes the hierarchy's main trade-off visible. When the global representation already places the correct reference near the top, a small shortlist retains the observed success of exhaustive learned matching. When viewpoint changes move the useful reference down the ranking, exhaustive search can recover cases missed by the primary hierarchy. The runtime comparison in Chapter 9 quantifies the cost of that additional search.

8.6.1 Unknown places and false acceptance

Table 8.7 reports the unknown-query diagnostic. It is possible for learned local matching to accept a geometrically plausible wrong candidate even when the image collection is disjoint. Increasing the candidate count exposes more such opportunities. A no-geometry policy necessarily accepts a top-ranked candidate for every unknown query in this implementation.

These failures matter for social AR because a false acceptance can display a post at the wrong physical location. The small diagnostic cannot establish a production false-positive bound, particularly for nearby look-alike buildings or adversarial replay. It does demonstrate why retrieval similarity and homography support cannot be treated as complete identity assurance. The

Table 8.7: Measured open-set diagnostic against a disjoint-collection gallery. Global-only ranking is omitted because ranking alone makes no acceptance decision.

Variant	Unknown queries	False acceptances	Rate (%)
A: SIFT	12	0	0.0
B: SP+LG	12	2	16.7
Full ($k = 1$)	12	0	0.0
D: Full ($k = 3$)	12	0	0.0
Full ($k = 5$)	12	1	8.3
No geometry	12	12	100.0

discussion of abuse in Chapter 10 follows directly from this measured limitation.

8.7 Short projection training

Table 8.8: Short frozen-backbone projection training, three fixed seeds. Rates are percentages. HPatches rows use only held-out place identities. Intervals are clipped to the feasible percentage range and are unstable with only three seeds.

Protocol	Descriptor	Mean R@1	SD	95% t interval
HP illum.	Pretrained 512	100.0	0.0	100.0–100.0
HP illum.	Head, no night aug.	100.0	0.0	100.0–100.0
HP illum.	Head + night aug.	100.0	0.0	100.0–100.0
HP view.	Pretrained 512	77.5	0.0	77.5–77.5
HP view.	Head, no night aug.	75.8	5.2	62.9–88.8
HP view.	Head + night aug.	74.2	1.4	70.6–77.8
Tokyo D–D	Pretrained 512	41.7	0.0	41.7–41.7
Tokyo D–D	Head, no night aug.	30.6	2.4	24.6–36.5
Tokyo D–D	Head + night aug.	37.5	7.2	19.6–55.4
Tokyo D–N	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo D–N	Head, no night aug.	94.4	6.4	78.6–100.0
Tokyo D–N	Head + night aug.	94.4	4.8	82.5–100.0
Tokyo D–S	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo D–S	Head, no night aug.	97.2	2.4	91.2–100.0
Tokyo D–S	Head + night aug.	98.6	2.4	92.6–100.0
Tokyo N–D	Pretrained 512	100.0	0.0	100.0–100.0
Tokyo N–D	Head, no night aug.	93.1	4.8	81.1–100.0
Tokyo N–D	Head + night aug.	94.4	4.8	82.5–100.0

Table 8.8 compares the pretrained descriptor with the short projection-head experiments on identical held-out queries. The main outcome is that the selected adaptation does not establish a consistent improvement over the pretrained representation. Some individual seeds recover or improve particular held-out cases, but that is insufficient reason to replace the original descriptor in the primary pipeline.

The reported intervals use only three fixed seeds and are therefore broad where the outcomes vary. Deterministic pretrained scores repeated across seed rows have zero seed variance, which does not mean zero data uncertainty. The table separates training randomness from the unmeasured variability of collecting different places. No best-seed checkpoint was selected for the deployment claim.

8.8 Qualitative correspondences and overlays

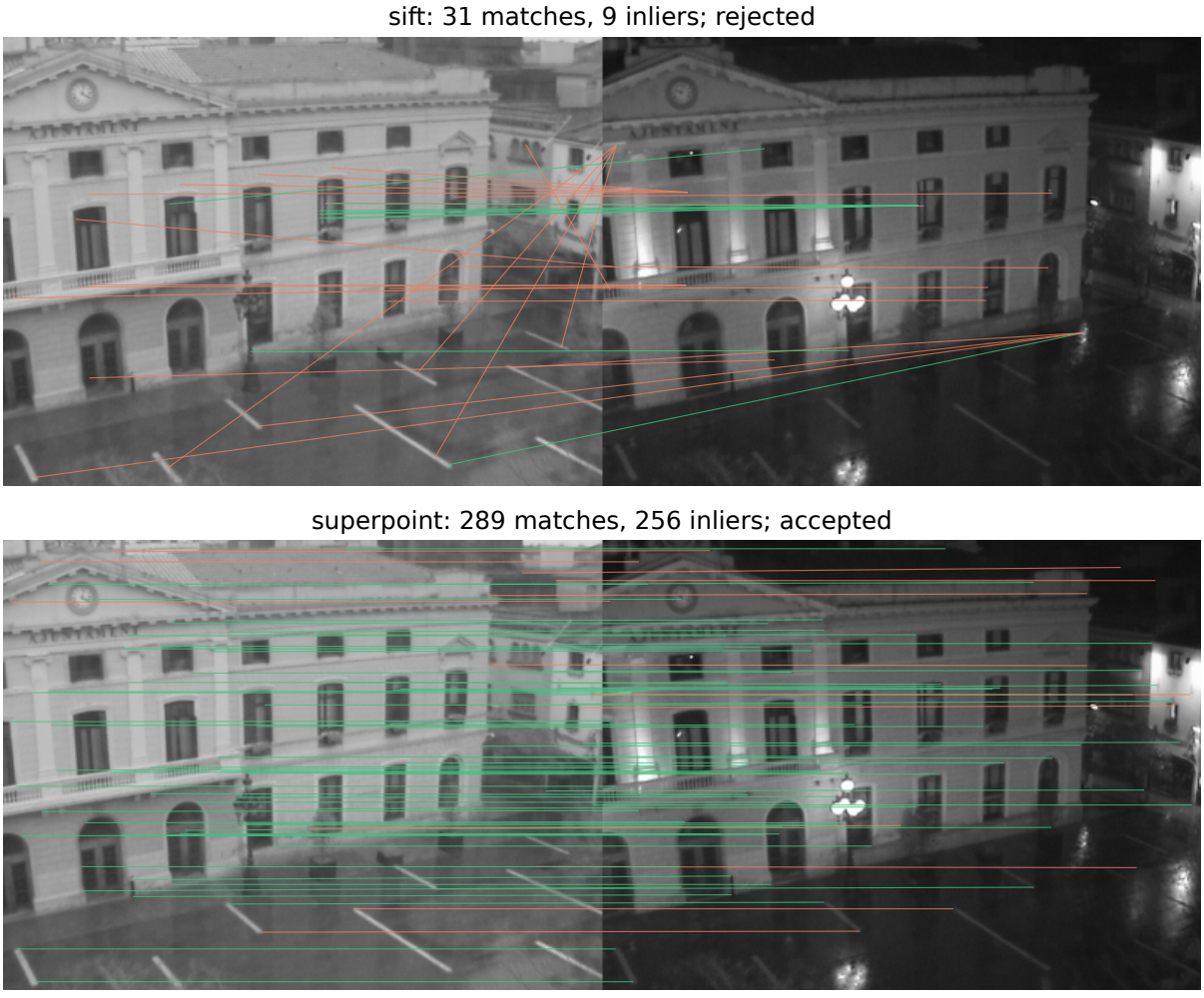


Figure 8.4: Executed correspondence visualisation for the documented illumination example. At most eighty actual tentative correspondences are drawn per panel; green denotes the saved RANSAC inlier mask and orange denotes other tentative matches.

Figure 8.4 shows how the numerical support arises on a selected example. The displayed lines are a deterministic subset of actual correspondences, not an illustration invented independently of inference. Their colour reflects RANSAC compatibility, which should not be confused with an independent manual label for every match.

Figure 8.5 links the geometry to the original application. An accepted map can support an

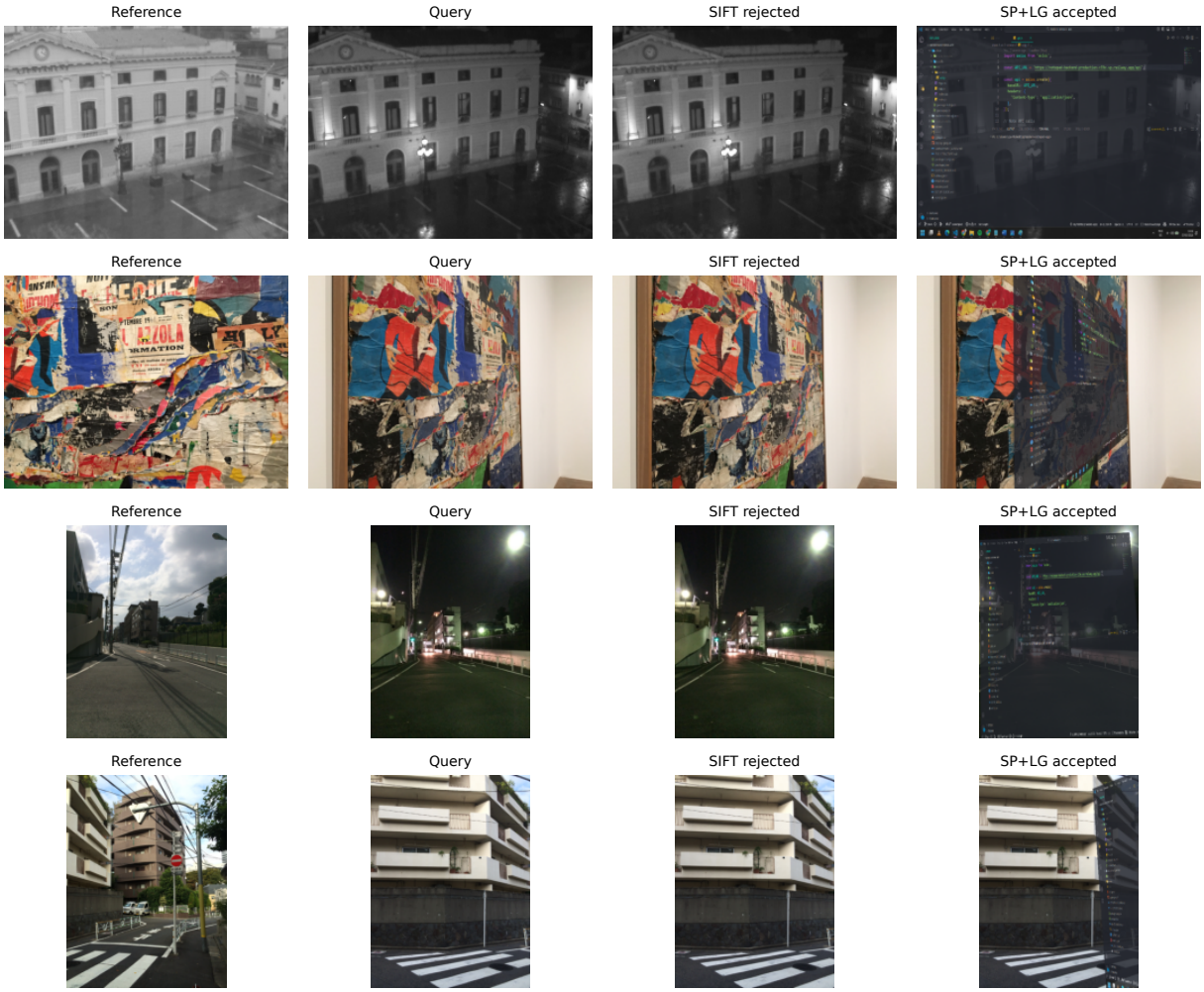


Figure 8.5: Saved-transform overlay examples across illumination, viewpoint and Tokyo protocols. Rejected geometry leaves the query unchanged. Accepted Tokyo overlays remain planar demonstrations without ground-truth pose or full-scene planarity.

attached image, and rejection remains visible rather than being replaced with a ground-truth or manually aligned overlay. A plausible street overlay is still not proof of correct depth, occlusion, or world-scale persistence.

The demo uses real selected day and night photographs with generated small two-dimensional camera motion. It is explicitly labelled as generated motion rather than real temporal video captured in those conditions. Both complete pipelines execute on each generated query frame. The bundled source video is separately used for the original-script execution and CLI smoke test. These demonstrations establish implemented data flow and rendering, not high-frequency real-world tracking performance.

8.9 Uncertainty and unavailable measurements

Table 8.9: Finite-sample uncertainty for correctly verified places. These intervals describe this selected set under an independent-site approximation; they do not compensate for geographic selection bias.

Protocol	Method	Correct	95% Wilson interval (%)
Tokyo D–D	A: SIFT	10/24	24.5–61.2
Tokyo D–D	B: SP+LG	13/24	35.1–72.1
Tokyo D–D	D: Full ($k = 3$)	9/24	21.2–57.3
Tokyo D–N	A: SIFT	8/24	18.0–53.3
Tokyo D–N	B: SP+LG	24/24	86.2–100.0
Tokyo D–N	D: Full ($k = 3$)	24/24	86.2–100.0
Tokyo N–D	A: SIFT	8/24	18.0–53.3
Tokyo N–D	B: SP+LG	24/24	86.2–100.0
Tokyo N–D	D: Full ($k = 3$)	24/24	86.2–100.0

Table 8.9 prevents perfect observed rates from being read as certainty. The Wilson intervals remain finite-sample statements about the chosen site set under the stated approximation. Cluster-bootstrap intervals for HPatches are also retained in the generated statistics artifact. Neither procedure removes the selection bias of a small geographic subset or repairs missing calibration.

Table 8.10: Metric localisation results are unavailable, rather than zero. Neither GPS site labels nor pixel homographies establish these errors.

Requested pose criterion	Measured rate	Reason
$(0.25\,\mathrm{m}, 2°)$	—	No calibrated query pose/map
$(0.5\,\mathrm{m}, 5°)$	—	No calibrated query pose/map
$(5\,\mathrm{m}, 10°)$	—	No calibrated query pose/map

Table 8.10 records a requested but unmeasured result. No metric translation-and-rotation rates are inferred from GPS identities or homography corner errors. Likewise, there is no real seasonal benchmark result, phone power measurement or city-scale service cost. The report

distinguishes these absences from actual negative results such as wrong accepted places or reduced retrieval after adaptation.

8.10 Threats to validity

8.10.1 Internal validity

The normalised local comparison shares preprocessing dimensions, feature budgets and geometry settings, but it cannot equalise the internal training data or detector semantics of pretrained models. Approximate descriptor search and floating-point kernels can introduce small run differences. Fixed seeds, exact model hashes and cached pair accounting reduce ambiguity, while the separate original-script runs avoid conflating the refactor with unchanged upstream behaviour.

Some evaluation stage timings were recorded while independent preparation tasks were also active. They are therefore treated as accounting diagnostics. The isolated complete-query profiles are the principal latency evidence. Even those profiles use a small fixed query selection and a warmed process, so they do not describe cold mobile startup or worst-case thermally constrained execution.

8.10.2 Construct validity

Inlier support is not ground-truth correspondence accuracy, accepted geometry is not true alignment, and place recognition is not metric localisation. The thesis reports those constructs separately. The planar corner metric itself measures extrapolation to the reference boundary, which may penalise a locally useful surface match. That is appropriate for a full-reference overlay but should be reconsidered for a smaller user-defined anchor region.

The custom Tokyo gallery changes the benchmark problem. Its perfect aligned cross-time results should be read alongside its viewpoint failures and small denominator. Synthetic night and fog labels describe transformations, not natural capture conditions. The proposed mobile architecture is not an implemented phone deployment. These boundaries are retained in captions and prose to prevent a demonstration from being used as evidence for a stronger construct.

8.10.3 External validity

The selected scenes, geographic region and reference directions do not cover the intended worldwide application. There are no measured seasonal transitions, construction changes, crowds, adversarial displays or long-term map revisions. A production gallery would include many more near-duplicate structures and unknown places. Extrapolating the measured false-acceptance rate or retrieval saturation to that setting would be unjustified.

8.10.4 Statistical conclusion validity

The sample is small and correlated. Three training seeds cannot characterise the full optimisation distribution, and a larger number of synthetic pixels does not create independent physical places.

Intervals are presented with their grouping assumptions, not as proof of a universal method ranking. The conclusions emphasise supported direction and failure mechanisms rather than claiming statistical superiority across all deployment conditions.

8.11 Implications for the research questions

The measured evidence supports learned local matching for the selected illumination changes, while providing no uniform viewpoint advantage under the planar criterion. Global retrieval can retain aligned cross-time recognition with fewer local comparisons, but it limits success under the tested direction change. Geometry reduces indiscriminate acceptance yet still produces some wrong-place decisions. Short projection adaptation is not a reliable improvement here. The executed export establishes a practical descriptor-only deployment artifact, whose trade-offs are examined next.

Chapter 9

Deployment on Mobile and Scalability

9.1 What the measured prototype establishes

The artifact executes a complete local pipeline, but its measured host behaviour does not support running full learned recognition on every display frame. The primary full pipeline has an isolated median query-and-render time of 1332.5 ms on the recorded CPU configuration. Exhaustive learned matching takes 3020.4 ms, exhaustive SIFT takes 186.2 ms, and global-only retrieval takes 18.5 ms. These generated values describe different functionality, so the fast global result cannot be substituted for a complete anchored query.

Table 9.1: Isolated warm CPU query and rendering measurements with cached references. RSS is the complete Python process peak, including imports, models, gallery and allocator state.

Pipeline	Median ms	95th pct. ms	Peak RSS MB
classical	186.2	211.3	568.3
local	3020.4	4040.9	1435.5
retrieval	18.5	21.4	537.2
full	1332.5	1523.1	1492.3

Table 9.1 includes upper-tail latency and process peak memory. The full learned path requires substantially more host resources than the classical recogniser. Reference caching reduces repeated work but also retains feature tensors. The process peak includes Python and library imports as well as weights and intermediate buffers; it is not a phone application memory estimate.

Figure 9.1 supports the main deployment conclusion: a useful mobile design should decouple occasional recognition from continuous native tracking. This is a design implication of the measured workload, not a claim that such a phone integration has been completed. The original tracking script's faster frame average in Table 3.1 illustrates the distinction between finding a target and maintaining an already acquired relation.

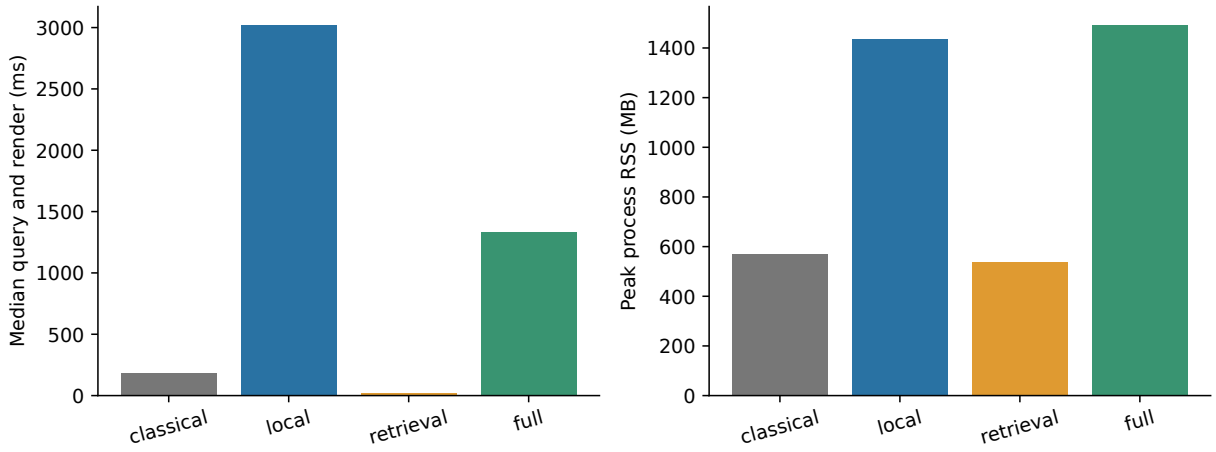


Figure 9.1: Measured isolated host query latency and complete-process peak RSS. Reference features are cached and query images are predecoded. The global-only path does not perform geometric anchoring.

9.2 Executed ONNX export

The CosPlace global model was exported to a fixed-input ONNX graph and checked with the ONNX validator. The graph uses the same learned weights and descriptor normalisation as native PyTorch. Float ONNX outputs were compared with native descriptors on the actual downloaded Tokyo images. The observed maximum absolute difference is recorded in Table 9.2; it is sufficiently small for the float retrieval rankings in this run to match the native results.

Table 9.2: Executed global-descriptor export and quantisation on the host CPU. Parity is measured against native PyTorch across downloaded Tokyo images. Timing excludes preprocessing and is not a phone measurement.

ONNX variant	MB	Median ms	95th pct. ms	Max abs. error
fp32	45.75	24.04	29.71	5.5e-07
int8	11.52	17.32	25.79	0.07

The export scope is deliberately narrow: it covers the global descriptor, not SuperPoint, LightGlue, RANSAC, rendering, or a complete camera application. A successful descriptor graph does not establish that dynamic keypoint operations and adaptive matching export without changes. Nor does graph validation prove that a chosen mobile execution provider supports every operator efficiently. No TFLite, CoreML or phone-native full-pipeline artifact is claimed.

The executed float graph is 45.75 MB and has median descriptor inference of 24.04 ms in the recorded ONNX Runtime CPU test. This timing excludes image preprocessing and gallery search. It also differs from the separate native global query profile because the runtime, timed work and query population differ. The thesis retains these definitions rather than combining the faster number from one test with the broader functionality of another.

9.3 Executed INT8 quantisation

9.3.1 Calibration and graph form

Static quantisation used QDQ graph representation with a calibration reader drawn only from the reserved HPatches training places. Convolution and supported linear operators were selected for quantisation; remaining operations continue in floating point. The model is therefore a mixed graph with INT8-weight and activation paths, not a claim that every computation uses integer arithmetic. ONNX Runtime documents the distinction between graph representations and quantisation settings [24].

In a generic affine quantiser, a real value x is mapped to

$$q = \mathrm{clip}(\mathrm{round}(x/s) + z, q_{\min}, q_{\max}), \qquad \widehat{x} = s(q - z), \tag{9.1}$$

where s is a scale and z a zero point. Calibration estimates ranges for the selected activations. This introduces approximation error and possible saturation; a successful graph conversion alone cannot establish preservation of descriptor rankings. The artifact consequently performs actual quantised inference and retrieval evaluation.

9.3.2 Measured size, latency and accuracy

The executed quantised graph occupies 11.52 MB and has median descriptor inference of 17.32 ms in the recorded CPU test. Both reductions are measured from produced files and runtime calls. Descriptor parity error is larger than for the float graph, as expected from the approximation, but the practical consequence must be read from retrieval rather than from an arbitrary tolerance alone.

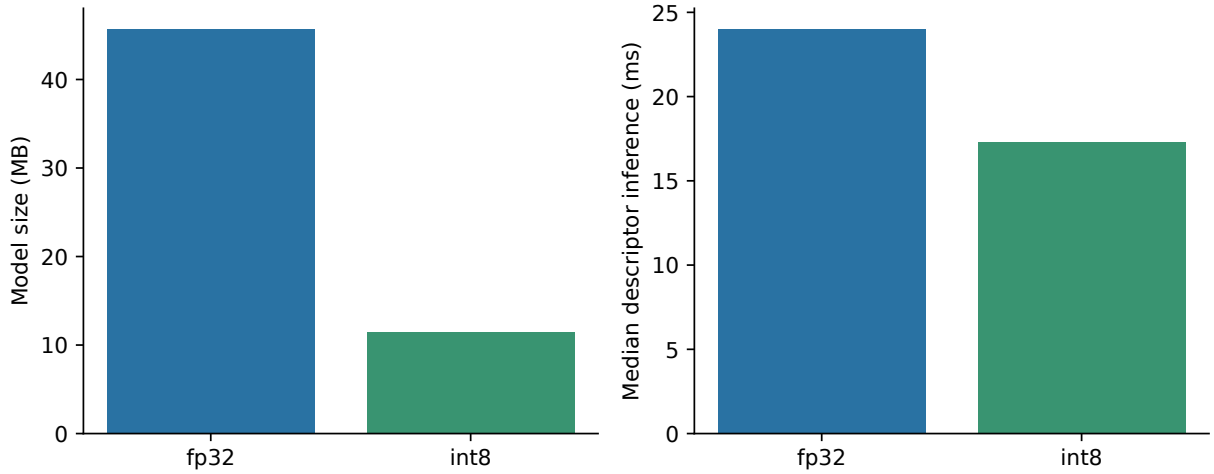


Figure 9.2: Actual global-descriptor ONNX file size and CPU inference after QDQ quantisation. These measurements exclude the local matcher and do not represent an end-to-end phone speedup.

Table 9.3: Retrieval accuracy (percent) after actual ONNX inference. Calibration uses training places, not Tokyo queries.

Protocol	Float R@1	INT8 R@1	Float R@5	INT8 R@5
tokyo_day_to_day	41.7	33.3	62.5	70.8
tokyo_day_to_night	100.0	100.0	100.0	100.0
tokyo_night_to_day	100.0	100.0	100.0	100.0
tokyo_day_to_sunset	100.0	100.0	100.0	100.0

Figure 9.2 and Table 9.3 show the trade-off. The aligned cross-time subsets retain their observed saturated retrieval, but the difficult daytime viewpoint split changes ranking accuracy. Thus quantisation is not accuracy-free even though several protocol scores remain unchanged. Calibration on training scenes also leaves an appearance-domain mismatch with Tokyo, which a broader deployment study should investigate.

The global encoder is only one part of the full query cost. Reducing it cannot remove the much larger local extraction and matching workload observed in the host profile. The descriptor export is still useful because it produces a concrete deployment candidate and reduces storage, but it does not establish that the full learned AR pipeline meets an interactive frame budget.

9.4 Recognition and tracking schedules

Design scope. The following specification describes an engineering extension; it is not a claim that a deployed multi-user service was measured.

A proposed mobile state machine has unlocalised, searching, verified and tracking states. Searching invokes retrieval and local verification. A verified observation establishes a relation to the stored place representation. A native visual-inertial tracker then propagates the camera pose at the platform's tracking rate. Loss of tracking, large uncertainty, or an elapsed refresh policy can trigger recognition again.

This design separates two failure types. Recognition failure means no supported map relation was found. Tracking failure means the relation was previously known but current device motion can no longer be estimated reliably. Reusing the last recognition output indefinitely would conceal the second failure. Instead, anchor visibility should depend on current tracking quality and the age of the last verified relation.

The state machine would require explicit time accounting. If a recognition call takes T_r and is requested at an interval Δ, the average recognition workload scales with T_r/Δ when calls do not overlap. This expression is a scheduling model, not a measured battery result. A real implementation must also account for camera processing, graphics, model warmup, thermal throttling and concurrent native tracking.

9.5 Reference coverage and GPS priors

The Tokyo viewpoint failure suggests a reference-coverage problem as well as a representation problem. A single front-facing reference may share too little visual content with a side-facing query. A proposed place model should therefore retain several directions or local submaps. Additional references raise storage and matching costs, making the retrieval hierarchy more important. This collection strategy remains a hypothesis until evaluated with an expanded gallery.

If the user permits a coarse location prior, candidate retrieval could first restrict the map region. The visual score could be combined with a spatial prior schematically as

$$S_i = \lambda s_i + (1 - \lambda) \log P(p_i \mid \ell), \tag{9.2}$$

where ℓ is the permitted location observation. The parameter λ and prior model would need validation. The current experiments do not use GPS filtering, so their results cannot be attributed to such a prior.

Location filtering creates its own failure mode when a noisy or denied position excludes the correct place. A mobile client should support uncertainty-aware region expansion or a visual-only fallback. It should also avoid logging detailed movement simply to make retrieval convenient. The privacy analysis treats the prior as an optional input with a distinct retention policy.

9.6 Deployment gates supported by the evidence

The present artifact is suitable for reproducible CPU research and a planar content demonstration. The descriptor-only ONNX graph and quantised variant are concrete export outputs. A phone pilot still requires native camera orientation handling, runtime-provider validation, full local-model deployment or a server split, calibrated map alignment, tracking recovery and device power testing. A public multi-user release additionally requires the access, moderation and location-privacy controls developed in the next chapter.

Chapter 10

Ethics, Privacy and Safety

10.1 Why the application changes the risk model

Stitch associates digital content with physical places and retrieves it through camera observations. That combination can reveal more than a conventional image-processing demonstration. A query may expose where a person is standing, which private space they are visiting, or which content they are permitted to see. A persistent post can also affect a person who never installed the application if it is attached to their home or workplace.

The present artifact is a local research prototype. It does not upload user camera frames to a service, publish posts, authenticate users or enforce production moderation. This boundary reduces the actions performed during the experiment but does not resolve the requirements of the proposed application. The following analysis treats those requirements as design obligations and identifies where the current software supplies no guarantee.

10.2 Captured faces, vehicles and private details

Reference and query images can include people, number plates, documents, screens and interiors. Even when these details are irrelevant to place recognition, they may remain in stored imagery or be recoverable from distributed assets. The research therefore uses documented public datasets and the supplied repository media rather than collecting new unconsented live footage. Dataset availability is not treated as proof that every depicted person's privacy concern has disappeared.

A deployment should minimise the imagery retained for localisation and review whether redaction is needed before storage or sharing. Face and number-plate filtering itself can fail, so an implementation should not label a map "anonymous" merely because a detector ran. Human review, a reporting channel and bounded retention may still be appropriate. No face or plate detector is evaluated in this artifact, and no redaction accuracy is claimed.

The map's visual requirements can conflict with privacy. A repeated window pattern or storefront sign may be useful for recognition while also identifying a sensitive place. Replacing pixels with descriptors can reduce direct image exposure but does not prove that the resulting representation contains no location or identity information. The thesis consequently avoids calling

local descriptors or global embeddings inherently anonymous.

10.3 Location privacy and retrieval logs

A sequence of recognised places can become a movement history. Such a history may be more sensitive than an individual query because it reveals routines and associations. An engineering design should therefore separate transient inference from persistent analytics. A client can use a place hypothesis to render authorised content without retaining a timestamped record of every unsuccessful or successful observation.

A GPS prior should be optional and purpose-limited. Its accuracy and retention policy should be separate from the camera-processing policy. If location permission is denied, the application should expose a constrained visual-only mode or explain the functional limitation without covertly reconstructing a movement log. The current benchmark uses site metadata for evaluation labels, not a live tracking service.

The public legislative text of the GDPR articulates purpose limitation, data minimisation and storage limitation among its principles [25]. These principles inform the proposed design; the thesis does not certify compliance in any jurisdiction. Deployment would require review of the actual data flows, roles, lawful basis and applicable local law. A research dataset licence and an application privacy notice address different questions and cannot substitute for one another.

10.4 Consent and control over persistent content

The social concept needs more than permission from the person creating a post. A virtual object attached to another person's property or image may affect that person and later visitors. The application should define who can create public anchors at a place, whether owners or community managers can request removal, and how private or restricted locations are handled. Technical visibility of a surface does not itself establish permission to use it as a public message board.

Posts should have explicit visibility and expiry rules. Private content should be retrieved only after authorisation, even if the place is correctly recognised. Shared spatial coordinates can support several audiences without exposing all content to every visitor. The current JSON schema illustrates author and content identifiers, but it contains no authentication or access-control engine. Those fields are examples, not security guarantees.

Deletion should cover more than a row in the post database. Cached assets, thumbnails, search indexes and map associations may retain the content or its location. A proposed service should define how those copies are invalidated and what minimal audit information remains. The file-based prototype makes no distributed deletion claim, and its research outputs should not be mistaken for a retention policy for production users.

10.5 Moderation and misuse of place-bound posts

Persistent AR can be used for harassment, impersonation, targeted advertising, offensive content or deceptive navigation cues. Attaching a message to a real building may imply endorsement by the building's occupants. Even a harmless image can be abusive when repeatedly placed at a particular person's home. Moderation must therefore consider the relation among content, target place, audience and author, not only a classifier score on the uploaded asset.

A proposed moderation flow should support reporting, temporary hiding, appeal and removal. Limits on repeated posting and duplicate anchors can reduce spam, but limits should not be described as protection against all abuse. Human escalation is likely to be required for context-dependent cases. The present artifact implements none of these workflows and does not send any posts to an external service.

The original "digital garbage" metaphor should also be treated carefully in product design. It motivates playful spatial content, but uncontrolled accumulation can obscure the real environment or overwhelm an intended public place. User-controlled density, visibility and content categories are therefore important interface choices. These are requirements inferred from the application's persistence and public-space setting, rather than measured user-experience findings.

10.6 False localisation as an application hazard

The measured open-set diagnostic in Table 8.7 shows that geometric support can still produce a wrong-place acceptance. Such an error can display content on an unintended facade or suggest that a message belongs to the current location. A recogniser should therefore communicate uncertainty and avoid presenting a weakly supported anchor as authoritative spatial guidance.

This concern becomes more serious when AR content resembles instructions, warnings, official signage or emergency information. The current prototype is an image-post demonstration and should not be treated as a navigation or safety system. A future product should restrict misleading presentation and preserve a clear distinction between application content and real-world authority. The thesis makes no claim that the existing verifier is sufficient for safety-critical use.

Visual replay is another unresolved threat. A photograph or display showing a known place can trigger appearance-based recognition without the device being physically present there. Neither a correct descriptor rank nor a homography alone proves physical presence. A stronger presence claim would require additional signals and a threat model, and those signals should be evaluated without creating disproportionate location surveillance.

10.7 Dataset and model licensing

The artifact separates imagery, software implementations, pretrained checkpoints and generated results. Each may carry different conditions. Official Aachen and RobotCar documentation, for example, specifies non-commercial academic restrictions; those datasets were not imported into the executed image set. HPatches includes source-specific attribution, and the Tokyo archive

should not be assumed to grant unrestricted commercial redistribution.

The learned implementations are obtained from their official repositories and model hashes are recorded. Retaining a repository licence is necessary but may not resolve all terms associated with third-party weights or source datasets. The thesis therefore supplies provenance and links rather than a blanket assertion that the resulting application is commercially cleared. A production release should review every dependency and data source actually distributed.

10.8 Responsible progression beyond the prototype

A limited pilot should first use a small consented set of places and non-sensitive content, with clear deletion and reporting mechanisms. It should measure wrong-place acceptance, tracking loss and content exposure alongside recognition recall. A later public system would need a broader place and device population, operational moderation and independent review of its data practices. The current evidence supports beginning that engineering evaluation; it does not support bypassing it.

Chapter 11

Conclusion and Future Work

11.1 Summary of the implemented work

This thesis converted Stitch's classical reference-overlay demonstration into a reproducible research artifact for place recognition and planar content anchoring. The original scripts were retained and executed through a bounded file-input adapter. A modular package adds global retrieval, learned local matching, shared geometric verification, explicit alpha rendering, a place/-content database, configuration and CLI operation. The report is built from saved predictions, data and checkpoint manifests, exact execution logs and generated figures and tables.

The evaluated data comprise 144 selected HPatches images with 120 ground-truth planar pairs, 96 real Tokyo photographs from 24 selected sites, and 90 seeded synthetic queries. These collections establish different evidence levels. The artifact reports true planar accuracy where homographies exist, place correctness where site labels exist, and unavailable metric pose rates where the necessary calibration and map are absent.

11.2 Answers to the research questions

11.2.1 Local correspondence under appearance change

The selected HPatches illumination comparison improves from 81.7% SIFT homography success to 100.0% with SuperPoint–LightGlue. This supports the use of pretrained local representations for the tested illumination changes. The viewpoint comparison yields 86.7% and 86.7%, respectively, and does not support a universal learned-method advantage. Geometric support and actual boundary accuracy must both be measured.

11.2.2 Global retrieval and candidate search

On the selected Tokyo day-to-night protocol, SIFT Recall@1 is 58.3%, while the learned local and CosPlace methods reach the observed saturated rate reported in the generated tables. The hierarchy correctly verifies 100.0% of those queries with a small shortlist. On the different-view daytime protocol, however, its correctly verified rate is 37.5%, showing that the global shortlist

can discard references recoverable by exhaustive local search.

The answer is therefore conditional: global retrieval is an effective search reduction when the useful view remains near the top of the global ranking. It is also an accuracy bottleneck when scene overlap or viewpoint reduces that ranking quality. The small custom Tokyo gallery and aligned cross-time views prevent direct comparison with official city-scale benchmark claims.

11.2.3 Verification and anchoring

Retrieval success is insufficient evidence for rendering. SIFT correctly verifies 33.3% of the selected day-to-night queries despite a higher retrieval recall, illustrating the difference between a ranked identity and accepted geometric support. The open-set diagnostic further shows that local verification can still accept a wrong candidate. The implemented rejection path is necessary, but a production identity or physical-presence guarantee requires stronger evaluation and controls.

The planar pipeline successfully connects a verified place record to its stored content and a rendered image overlay. It does not establish arbitrary three-dimensional persistence, occlusion correctness, world-scale pose or cross-device anchor consistency. Those require calibrated mapping and a native tracking integration beyond the current image-to-image homography model.

11.2.4 Short adaptation

The executed projection-head training does not provide a consistent held-out improvement over the pretrained global descriptor. Three fixed seeds expose variability, and synthetic night augmentation does not remove the mismatch between the limited training scenes and the evaluation conditions. The pretrained descriptor remains the primary pipeline choice. This negative result narrows the roadmap: broader data and a compression control are needed before attributing improvement to appearance adaptation.

11.2.5 Deployment trade-offs

The descriptor-only ONNX export and QDQ quantisation both execute. File size changes from 45.75 MB to 11.52 MB, and the recorded ONNX CPU descriptor latency changes from 24.04 ms to 17.32 ms. The difficult viewpoint split loses some retrieval accuracy, so quantisation must be validated through task outcomes as well as file size.

The complete learned pipeline's isolated host median is 1332.5 ms. It improves on exhaustive learned search but does not support per-display-frame recognition on this configuration. A mobile design should use occasional recognition to establish a map relation and native tracking to maintain it. No phone runtime, thermal or power result is claimed.

11.3 Principal limitations

The strongest limitation is coverage. The gallery is small, geographic diversity is narrow, and real seasonal/weather transitions were not evaluated. Synthetic transformations cannot replace those

observations. The optional training uses very few place identities, and its seed intervals do not represent the uncertainty of collecting a new city or device population.

The second limitation is geometric scope. A homography is appropriate for a plane-induced map, not a complete street scene. Tokyo supplies no independent homography or calibrated pose for the selected pairs. An accepted overlay there is consequently a demonstration of the implemented decision, not a quantified metric-localisation result. The requested translation-and-rotation thresholds remain unavailable.

The third limitation is deployment completeness. The package is a local CPU research implementation, with a compact on-disk database and image-post renderer. It has no authenticated multi-user backend, distributed deletion, map revision service or operational moderation. Export covers the global descriptor only. The proposed service and mobile sections define requirements rather than claim an existing product.

11.4 Staged future work

The first stage should broaden evaluation before changing model complexity. A consented multi-direction reference set and larger public galleries can test the observed viewpoint bottleneck. RobotCar, Extended CMU and MSLS can add genuine seasonal and weather transitions. Their acquisition and label conventions should be frozen before inference, following the same manifest and command-log discipline used here.

The second stage should add calibrated map-relative localisation. A sparse reconstruction with known intrinsics and validated 2D–3D associations would support PnP and the requested metric thresholds. Anchor regions should be represented in map or plane coordinates, with uncertainty and explicit conventions. This stage should evaluate boundary error, reprojection support and actual camera pose separately.

The third stage should establish native mobile feasibility. It should export or otherwise deploy the local feature and matching stages, profile their supported backend operators, and measure complete camera-to-render latency on representative devices. Recognition scheduling, tracking drift, loss recovery, memory pressure, thermal behaviour and power should be measured under sustained operation rather than inferred from a short warmed host run.

The fourth stage should address persistent social operation. Place/map revisions, access-controlled content, consent, deletion and moderation must be implemented together. A pilot should measure false content placement and user-visible failure behaviour, not only retrieval recall. The open-set results show why a visually plausible map cannot be treated as complete authorisation or identity evidence.

11.5 Closing assessment

The work provides a concrete, auditable path from Stitch's supplied demonstration to a learned planar-AR research system. Its contribution lies in executed integration, controlled comparison

and honest boundaries. The evidence supports improved illumination correspondence and useful coarse retrieval on the selected data, while exposing viewpoint, false-acceptance, adaptation and runtime limitations that must shape the next implementation.

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Appendix A

Reproduction Instructions

A.1 Environment and setup

All commands below are issued from the delivered project root unless a directory change is shown. The setup script selects a supported CPython interpreter, creates a local virtual environment, installs the pinned requirements and local package, obtains the official learned-matcher source at its recorded revision, and downloads the pretrained checkpoints. The data directory, model directory and environment are excluded from source control.

```
bash scripts/setup.sh
make data
make baseline
make evaluate
make train
make export
make demo
make profile
make test
make thesis
```

The trained-head and export steps use the frozen data preparation. The demonstration and artifact generator consume the main evaluation results. The complete-query profile is executed after other heavy inference tasks finish. The output PDF is `report/thesis.pdf`; comparison clips and animation are in `demo/`. Exact arguments used in this run are preserved in `results/commands.jsonl`.

A.2 Direct CLI checks

```
.venv/bin/python scripts/run_logged.py .venv/bin/stitch index data/places.json
.venv/bin/python scripts/run_logged.py .venv/bin/stitch query data/places.json test.jpg results/raw/
    cli_query.png
.venv/bin/python scripts/run_logged.py .venv/bin/stitch video data/places.json video.mp4 results/raw/
    cli_video.mp4 --limit 8
.venv/bin/python scripts/run_logged.py .venv/bin/python scripts/download_data.py --verify-only
```

The query image is already represented in this demonstration database, so this CLI check is a functional content-retrieval check rather than a held-out accuracy result. The video command exercises decoding, EOF/limit handling, recognition and writing. Bounded file inputs replace the unavailable private DroidCam stream; no live phone capture is claimed.

A.3 Data integrity and restoration

`results/data_manifest.json` pins downloaded filenames, lengths and SHA256 hashes. `results/preparation.json` pins generated synthetic inputs and the frozen benchmark manifest. Existing valid collections are verified and skipped by the downloader. Missing collections can be restored from the recorded public sources and checked against the supplied manifest. A corrupt existing file fails explicitly, preventing unnoticed input replacement.

The original HPatches host and author mirror are documented in Chapter 5. Tokyo subset members are verified against ZIP CRC and decompressed size before their local hashes are recorded. Full-archive transfer failures remain in the command journal. Reproduction depends on the continued availability and terms of the original sources; their presence in this artifact is not a promise that a remote URL will remain unchanged.

A.4 Build procedure

The installed TeX toolchain was used directly. The document uses the report class, numeric natbib citations, generated vector figures and a nomenclature index. From the project root, `make thesis` regenerates empirical artifacts and invokes latexmk. The thesis Makefile copies the completed PDF to the main deliverable path and runs the verification script. A clean TeX rebuild is available through the recorded clean/build commands.

Authoring and environment-inspection commands performed before the command runner existed are summarised in `results/bootstrap_commands.json`. Subsequent substantive executions are recorded in `results/commands.jsonl` with exact command text and outcomes. File edits are represented by the delivered source, not presented as experiment commands.